NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANG TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	24-12-25	Issued For Review	RA	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

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Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	Company's Comment	Contractor's Response	Remarks

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1. GENERAL

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².

ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	967376,210	10,414	
PM-8593	425000,093	907528,777	19,046	

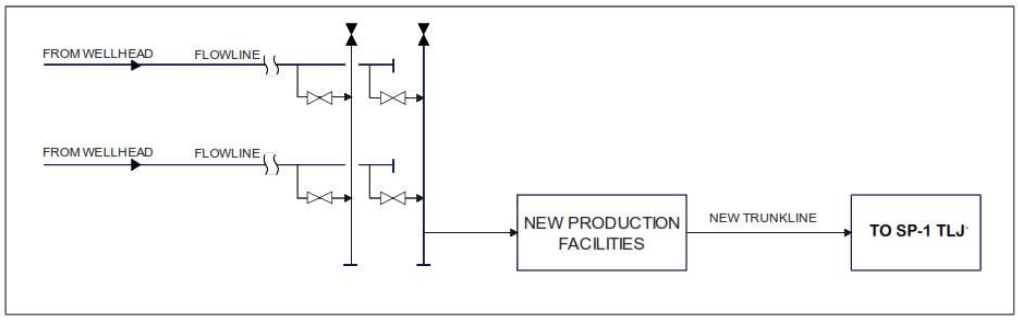
Gambar 1. Lokasi Rencana SP Tanjung Miring Barat - Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari flowline, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah idle atau tidak beroperasi, sehingga produksi sumuran menggunakan Tank On-Site (TOS) dan dikirim secara gross ke SP-II TMT menggunakan trucking. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan trucking. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan

spesifikasi yang dipersyaratkan di titik serah.

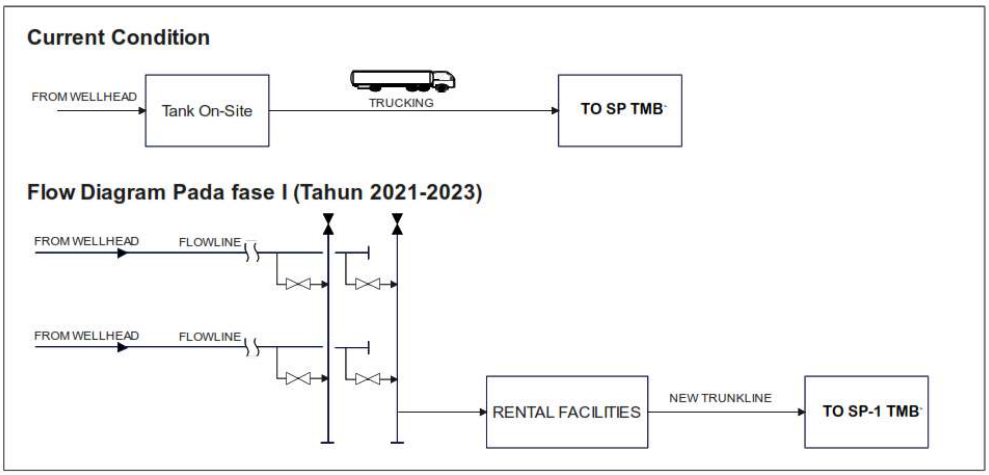
Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa Early Production Facility (EPF) untuk selanjutnya dikirimkan menuju SP-1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah flow/ine dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.



Gambar 2. Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 3. Rencana Pola Operasi Fase

1.2 Tujuan

Studi ini bertujuan untuk mengidentifikasi dispersi dan radiasi yang dihasilkan dari proses pembakaran gas buangan menuju Flare Stack FL-00 dalam tahapan desain lingkup pekerjaan proyek Pembangunan Fasilitas Produksi Stasiun Pengumpul (SP) Tanjung Miring Barat & Karangan Timur.

1.3 Ruang Lingkup

Studi ini mencakup analisis risiko dispersi dan radiasi yang dihasilkan dari proses pembakaran gas buangan menuju Flare FL-001. Studi ini berfokus pada peristiwa pelepasan hidrokarbon dan radiasi yang tidak diinginkan dari equipment yang terdapat di Flare FL-001, yang dapat menyebabkan kebakaran dan/atau ledakan, serta potensi kerusakan structural.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR.
SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa FEED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.
Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

2. KODE, STANDAR, DAN REFERENSI

- **STANDAR INTERNASIONAL**

No	Document Number	Title
1.	API-RP-752	American Petroleum institutes, Management of Hazards Associated with Location of Process Plant Permanent Buildings, 2009
2.	API 521	Pressure-relieving and Depressuring Systems
3.	-	"Vulnerability of humans" Risk Assessment Data Directory, International Association of Oil & Gas Producers, OGP. 434-14.1, March 2010
4.	CMPT	A Guide to Quantitative Risk Assessment for offshore installations, 1999

- **Dokumen Projek**

No	Document Number	Title
5.	TMBP-01-PR-PID-D115	P&ID FLARE SYSTEM
6.	TMBP-SP001-PR-HMB-D101	HMB FOR OVERALL FACILITY
7.	TMBP-SP001-PR-PFD-D101	PROCESS FLOW DIAGRAM FOR PRODUCTION MANIFOLD AND SEPARATION SYSTEM
8.	TMBP-01-PLP-D101	OVERALL PLOT PLAN
9.	TMBP-01-PR-DBS-101	PROCESS DESIGN BASIS DESIGN CRITERIA

- **Referensi Lain**

No	Document Number	Title
10.	TMBP-01-ME-DTS-155	MDS FLARE PACKAGE (FL-001)

3. Daftar Singkatan

Beberapa singkatan yang paling umum digunakan tercantum dalam daftar berikut:

ALARP	*As Low As Reasonably Practicable*
FEED	*Front End Engineering Design*
FS	*Flare Stack*
KRG	*Karangan*
MMSCFD	*Million Standard Cubic Feet per Day*
NPS	*Nominal Pipe Size*

PB	*Production Battery*
PT	*Perseroan Terbatas*
RP	Receptor Point
SP	*Stasiun Pengumpul*
SCH	*Schedule*
TMB	*Tanjung Miring Barat*
TMT	*Tanjung Miring Timur*
UFL	Upper Flammable Limit
WK	*Wilayah Kerja*

4. Metododologi

Perhitungan radiasi dan permodelan scenario dispersi jika terdapat kondisi pelepasan gas tidak terbakar langsung ke udara dilakukan menggunakan Flaresim, metode kalkulasi Flaresim API.

5. Konsiderasi Desain *Layout* dan Proses

Kondisi operasi yang akan menjadi skenario flaring pada Flare Stack mengacu pada Dokumen Heat and Material Balance [Ref.6] di cantumkan dalam tabel di bawah. Skenario flaring yang dilakukan dalam studi ini adalah sebagai berikut:

5.1 *Flaring Scenario*

Scenario berdasarkan jumlah gas berdasarkan kapasitas Flare Stack (FL-001). Dalam kasus ini, komposisi gas dan flowrate didapatkan dari aliran nomor 36 [Ref.6] dan kapasitas maksimum operasional Flare Stack (FL-001).

- Case 1 : Normal Operasi
- Case 2 : Kapasitas Design

5.2 Data Fluida

Data kondisi operasi fluida dan komposisi gas menuju flare mengacu pada Heat and Material Balance aliran nomor 36 [Ref.6] *tercantum pada tabel di bawah:*

Tabel 5.1 Data Fluida

Paramater	**Skenario Flaring**
Flowrate (MMSCFD)	2.83
Fluid Temperature (°F)	186.95
Ref. Pressure (psig)	3.59
Molecular Weight	25.85
Komposisi	
Methane	0.47
Ethane	0.08
Propane	0.14
i-Butane	0.02
n-Butane	0.03
i-Pentane	0.01
n-Pentane	0.01
n-Hexane	0.01
CO_2	0.03
Nitrogen	0.00
Oksigen	0.00
H_2O	0.20
Total	1.0000

5.3 *Data Flare Stack*

Stack dan tip data sebagai berikut [Ref.8]:

- Design Flowrate: 3.52 MMSCFD
- Diameter Stack : 12/6 inch
- Tinggi Flare : 82 ft/25 m (Tinggi total termasuk *tip*), 40 ft/12 m (tinggi tip)
- Setpoint tekanan menuju flare : 2.83 psig

5.4 Data Lingkungan

Data lingkungan diperloeh berdasarkan Process Design Basis Design Criteria [Ref.9]. Permodelan dibuat berdasarkan kondisi cuaca terburuk yang dicantumkan dalam tabel berikut:

Tabel 5.2 Data Lingkungan

Parameter	Unit	Min	Avg	Max
Temperature	°F	73.4	80.6	95
Humidity	%	87%		
Windspeed	m/s	17.88		
Solar Radiation	kW/m^2	1.04 (Note 1)		
Surface Roughness	m	0.25 (Note 2)		

Note:
1. Berdasarkan API 521 [Ref.2] untuk daerah tropis
2. Parameter untuk daerah tangki yang tersebar

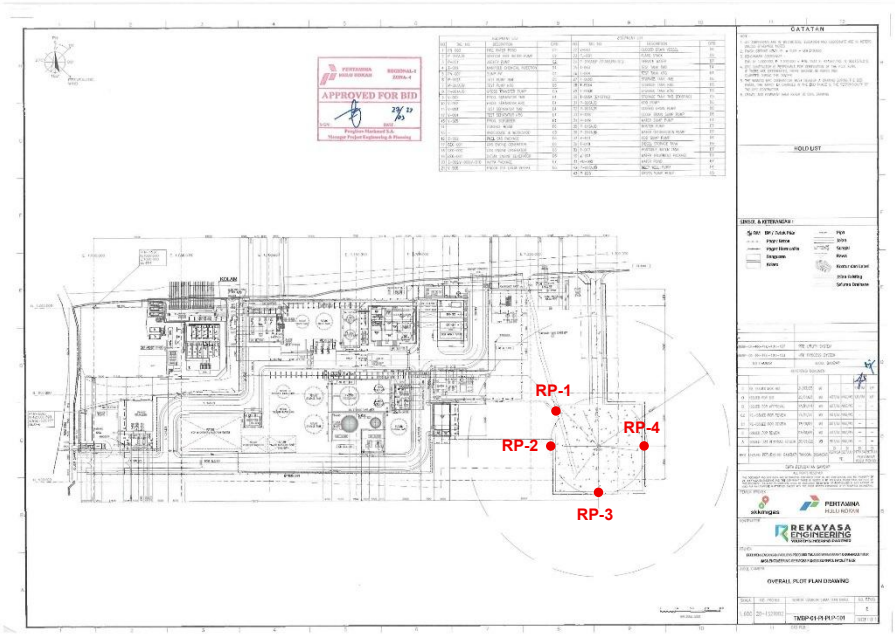
5.5 Data Receptor Point

Tabel 5.3 dan Gambar 5.1 menunjukkan data dan lokasi untuk setiap *receptor point* untuk Flare Stack (FL-001).

Tabel 5.3 Data Receptor Point Flare Stack (FL-001)

ID	Receptor Point	Northing, m (ft) (Note1)	Easting, m (ft) (Note1)
RP 1	Walkway	19.25 (63.15)	-25.01 (-82.04)
RP 2	Sistem perpipaan bagian barat Flare Stack (FL-001)	0.00 (0.00)	-26.52 (-87.01)
RP 3	Sistem perpipaan bagian selatan Flare Stack (FL-001)	-27.50 (-90.22)	0.00 (0.00)
RP 4	Sistem perpipaan bagian timur Flare Stack (FL-001)	0.00 (0.00)	26.49 (86.90)

Note:
1. Koordinat (0,0) mengacu terhadap posisi Flare Stack FL-001



Tabel 5.1 Lokasi Receptor Point Flare Stack (FL-001)

5.6 Kriteria Radiasi

Kriteria radiasi pada setiap receptor point mengacu pada API 521 [Ref.2], diantaranya:

a. Radiasi 2000 Btu/jam/ft^2 (6.3 kW/m^2) merupakan batas paparan terhadap personel selama 30 detik dengan pakaian kerja lengkap. Kriteria ini digunakan pada flare stack base saat emergency flaring.

b. Radiasi 1500 Btu/jam/ft^2 (4.73 kW/m^2) merupakan batas paparan terhadap personel selama 2-3 menit dengan pakaian kerja lengkap. Kriteria ini digunakan pada receptor point didekat flare stack saat emergency flaring.

c. Radiasi 500 Btu/jam/ft^2 (1.58 kW/m^2) merupakan batas paparan terhadap personel secara terus menerus dengan pakaian kerja lengkap. Kriteria ini digunakan pada semua receptor point (semua lokasi) untuk continuous flaring

Tabel 5.4 Kriteria Radiasi Flare Stack (FL-001)

ID	Receptor Point	Kriteria Emergency (Btu/h.ft^2)	Kriteria Continuous (Btu/h.ft^2)
RP 1	Walkway	1500	500
RP 2	Sistem perpipaan bagian barat Flare Stack (FL-001)	1500	500
RP 3	Sistem perpipaan bagian selatan Flare	1500	500

ID	Receptor Point	Kriteria Emergency (Btu/h.ft^2)	Kriteria Continuous (Btu/h.ft^2)
	Stack (FL-001)		
RP 4	Sistem perpipaan bagian timur Flare Stack (FL-001)	1500	500

5.7 Kriteria Dispersi

Kriteria dispersi untuk flammable gas mengacu pada kriteria berikut:

1. Konsentrasi 20% dan 40% LFL dimana merupakan kriteria aktivasi alarm
2. Konsentrasi 100% LFL dimana merupakan batas konsentrasi flammable gas mulai dapat terbakar di udara atmosfer

6. Hasil *Modelling* dan Analisis

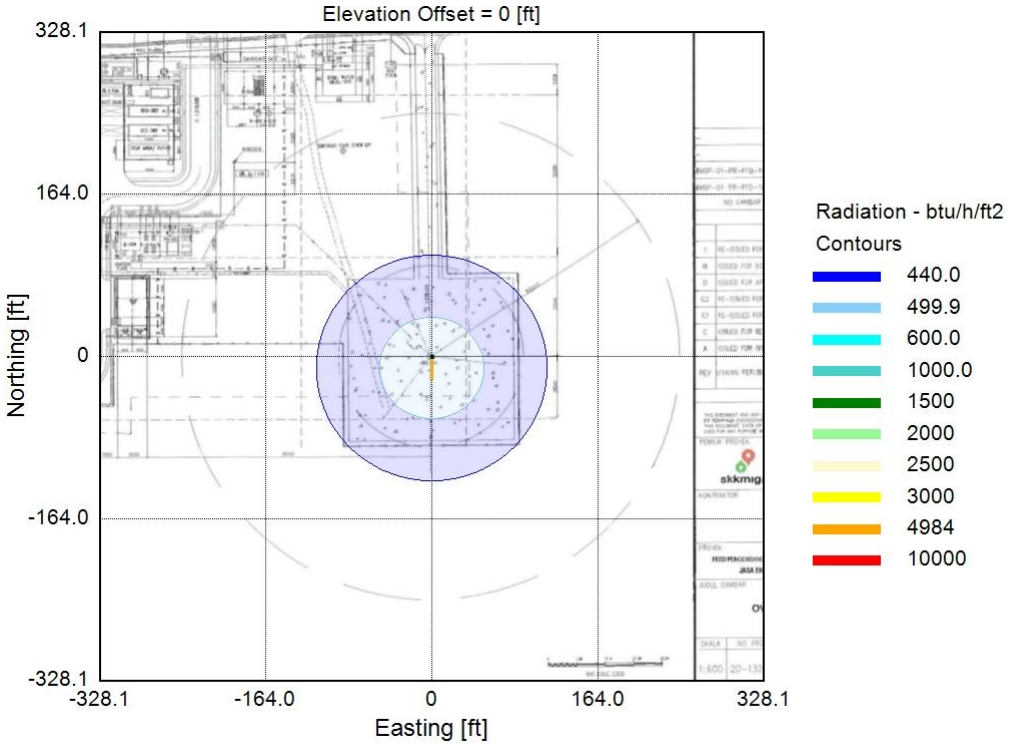
6.1 Radiasi Flare Stack

Simulasi flaring dilakukan untuk mengevaluasi intensitas radiasi yang diterima pada setiap receptor point. Hasil perhitungan penyebaran radiasi untuk seluruh skenario flaring pada Flare Stack FL-001 ditunjukkan pada Tabel 6.1 dan Gambar 6.1 s/d Gambar 6.2. Kriteria radiasi yang digunakan mengacu pada kondisi emergency, karena kondisi tersebut merepresentasikan skenario terburuk (worst-case) dari proses pembakaran gas menuju Flare Stack FL-001.

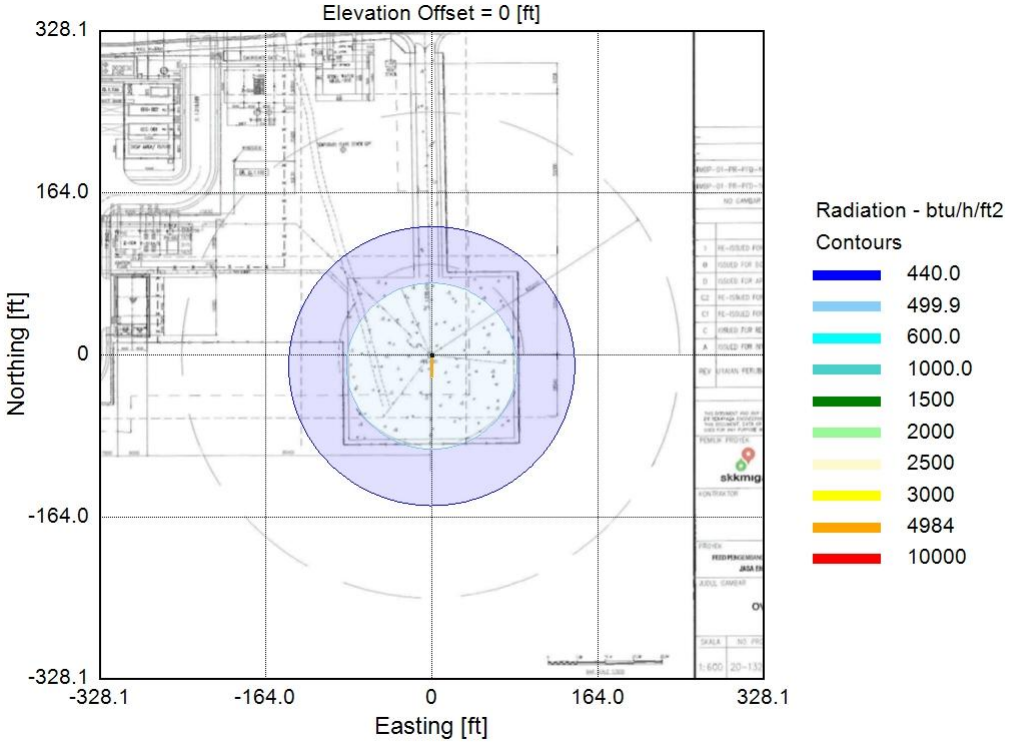
Tabel 6.1 Radiasi Flare Stack (FL-001)

ID	Receptor Point	Kriteria Radiasi Emergency (Btu/h.ft^2)	Radiasi Termal (Btu/h.ft^2)	
			Case 1	Case 2
RP 1	Walkway	1500	448	476
RP 2	Sistem perpipaan bagian barat Flare Stack (FL-001)	1500	472	505
RP 3	Sistem perpipaan bagian selatan Flare	1500	482	517

ID	Receptor Point	Kriteria Radiasi Emergency (Btu/h.ft^2)	Radiasi Termal (Btu/h.ft^2)	
			Case 1	Case 2
	Stack (FL-001)			
RP 4	Sistem perpipaan bagian timur Flare Stack (FL-001)	1500	472	505



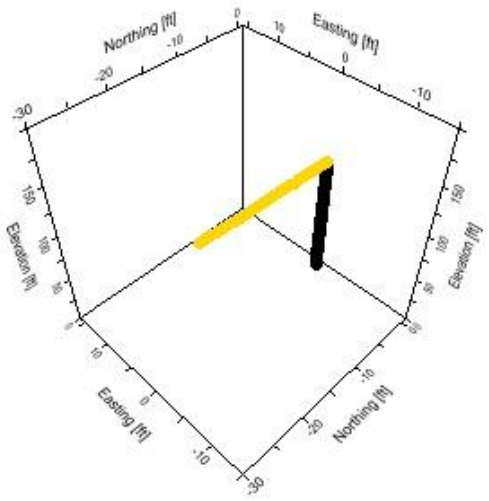
Gambar 6.1 Kontur Radiasi Flaring (FL-001) Case 1



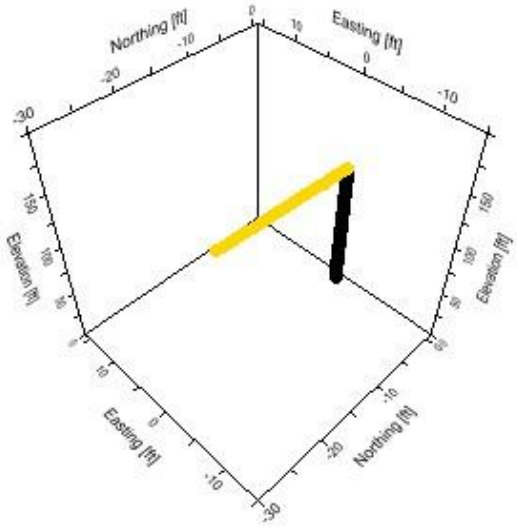
Gambar 6.2 Kontur Radiasi Flaring (FL-001) Case 2

6.2 Dispersi Flare Stack

Simulasi flaring dilakukan untuk mengevaluasi radius dan kontur dispersi dari gas keluaran Flare Stack (FL-001). Persebaran gas hasil pembakaran didapatkan sebagai berikut:



Gambar 6.3 Kontur Dispersi Flaring (FL-001) Case 1



Gambar 6.4 Kontur Radiasi Flaring (FL-001) Case 2

LAMPIRAN A
MODELLING FLARING

Environmental data

Environment: Environment 1 - Active

Atmosphere					
Pressure	14.696 psi	Temperature	95.0 F	Humidity	87.00 %
Wind Speed	58.661 ft/s	Wind Direction	0.00 °		
Background					
BackGround Noise	60.0 dB	Solar Radiation	330 btu/h/ft2		
Inc. Background Noise	Active	Inc. Solar Radiation	Active		
Transmissivity					
Method	Calculated	Min Value	1.00	Max Value	1.0
Wind Rose Calculations					
Wind Speed Array	Ignored	Wind Speed vs Direction	Ignored		
Dispersion Data					
Stability Class	PasquillD	Terrain Class	Urban	Surface Roughness	0.82
Wind Reference Height	32.81 ft	Correct Wind Speed	False		

Stack Configuration Data

Stack: FL-001 - Active

Input Data

Base point name	FL-001 (Base)	Base refers to	Origin		
Base - Northing	0.00 ft	Base - Easting	0.00 ft	Base - Elevation	0.00
Base - Angle from North	0.00 °	Base - Angle to Horizontal	0.00 °	Base - Radius	0.00
Stack					
Rotation from North	0.00 °	Angle to Horizontal	90.00 °	Stack Length	82.00
Sterile Area					
Elevation	0.00 ft	Noise Basis	Noise	Run SA Calcs	Fals

Fluid Data

Fluid: Fluid 1

Input Data

Thermo Method	Flaresim				
Use Temp Corr.	False	Use RK ZFactor	True		
LHV	16621.01 btu/lb	Tc	151.8 F	LEL	4.05 %
Fluid Temperature	187.0 F	Pc	1174.387 psi	Saturation	100.00 %
Molecular Weight	26.254	Cp/Cv	1.266	Ref. Pressure	18.286 ps

Composition in mole fraction

Methane	0.470000	Ethane	0.080000	Propane	0.14000
i-Butane	0.020000	n-Butane	0.030000	i-Pentane	0.01000
n-Pentane	0.010000	n-Hexane	0.010000	Water	0.20000
Carbon dioxide	0.030000	Nitrogen	0.000000	Oxygen	0.00000

Results

Flue to Feed Volume Ratio	3.820	Stoichiometric O2	2.500
Enthalpy @ 25°C	192.96 btu/lb	Enthalpy @ Fluid Temp.	243.66 btu/lb

Stoichiometric Flue Gas Composition

Carbon Dioxide	0.3769634 mole/mole	Water	0.6230366 mole/mole	Oxygen	0.0000000 mole/mol
Nitrogen	0.0000000 mole/mole	Sulphur Dioxide	0.0000000 mole/mole		

Tip Data

Tip: Tip 1 - Active

Input Data

General

Tip Type	Pipe	Located on stack	FL-001	Nbr of Burners	
Angle to Horizontal	90.00 °	Angle to North	0.00 °		
Tip Length	40.00 ft	Tip Diameter	6.00 in	Tip Opening	100.00 %
Seal Type	None	Diameter Basis	Tip		

Fluids

Fluid	Fluid 1	LP Fluid		Assist Fluid	
Fluid Mass Flow	8157.637 lb/h	LP Mass Flow	0.000 lb/h	Assist Mass Flow	0.000 lb/
Fluid Mole Flow	0.086 lbmole/s	LP Mole Flow	0.000 lbmole/s		

Pressure Drop

Riser Diameter	12.00 in	Tip Outlet Pressure	-----	Contraction Coeff.	1.00
Roughness	0.00098 in	Exit Loss	1.000		

Flame and Radiation

F Factor Method	User Specified	User F Factor	0.30	Correct User F Factor	Fals
Combustion Air Ratio	1.000	Flame Temperature	3631.7 F		
Flame Length Method	Brzustowski				
Radiation Method	Global	Flame Element Count	1	Element Position	50.00 %

Noise

Noise Method	Acoustic Eff.	Comb. Efficiency	0.00010 %	Peak Frequency	500 H
Jet Noise Method	Flaresim				

Purge Gas

Purge Gas Fluid	Nitrogen	Purge Area Basis	Stack	Purge Fixed Velocity	0.197 ft/s
Purge Fixed Volume Flow	0.003 ft3/s	Purge HUSA O2	6.00 %	Purge HUSA Height	25.00 ft

Emission Data

NOX Basis	Mass/Heat Release	CO Basis	Mass/Heat Release	HC Basis	Mass/Heat Releas
NOX Rate	0.0680320 lb/MMBtu	CO Rate	0.3700000 lb/MMBtu	HC Rate	0.1400000 lb/MMBt

Tip: Tip 1 - Results

General

Flame Length	26.04 ft	Heat Release	135.59 MMbtu/h	Tip End - Northing	0.00
API Flame Length	46.26 ft	Frac.Heat Radiated	0.30	Tip End - Easting	0.00
Exit Temperature	187.0 F	Actual Volume Flow	40.611 ft3/s	Tip End - Elevation	122.00
Assist Fluid Mass Flow	-----	Assist Fluid Flow Ratio	-----	Wind Speed At Tip	58.661 ft/

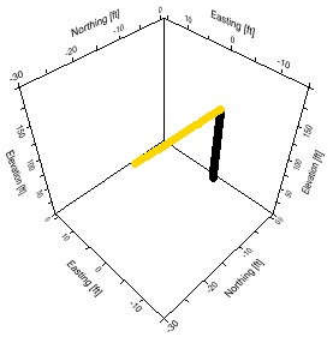
Velocities and Pressure Drop

Exit Velocity	206.828 ft/s	Tip Outlet pres.	14.696 psi	Total Outlet pres.	14.954 ps
Sonic Velocity	1242.661 ft/s	Tip Backpres.	14.997 psi	Total Backpres.	15.249 ps
Exit Mach Number	0.166	Seal Backpres.	15.093 psi	Total Backpres.	15.344 ps
Contraction Coeff.	1.000	Stack Backpres.	15.347 psi	Total Backpres.	15.362 ps
		Tip Pressure Drop	0.301 psi	Tip Total Pressure Drop	0.296 ps
		Seal Pressure Drop	0.096 psi	Seal Total Pressure Drop	0.095 ps
		Stack Pressure Drop	0.254 psi	Stack Total Pressure Drop	0.018 ps

Flame Shape Points

	X	Y	Z
1	0.00 ft	0.00 ft	122.00 ft
2	-23.47 ft	0.00 ft	128.81 ft

3D Flame Shape Chart



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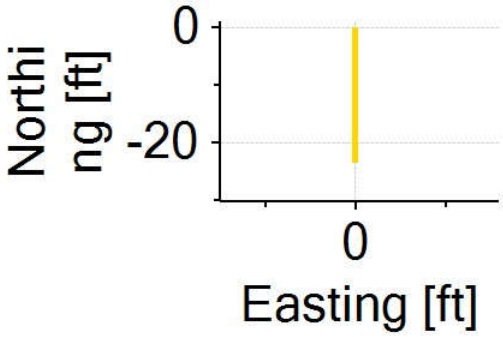
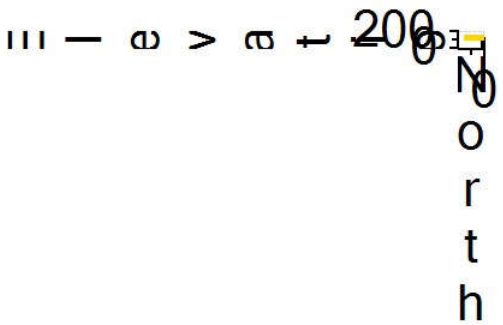
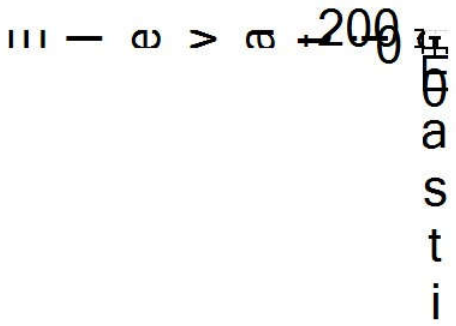
2D Flame Shape Chart, Northing-Easting



2D Flame Shape Chart, Northing-Elevation



2D Flame Shape Chart, Easting-Elevation



Noise Results

Peak sound pressure

Distance	98.43 ft	Sound pressure	95.5 dB

Noise Spectrum Results

Frequency	Total PWL	Combustion PWL	Jet PWL
Hz	dB	dB	dB
31.3	98.5	98.5	61.0
62.5	107.0	107.0	65.9
125.0	115.5	115.5	65.2
250.0	124.0	124.0	62.0
500.0	132.5	132.5	57.1
1000.0	129.7	129.7	51.5
2000.0	126.9	126.9	45.5
4000.0	124.1	124.1	39.5
8000.0	121.3	121.3	33.5
16000.0	118.5	118.5	27.5

Noise Spectrum

Combustion Results General					
Adiabatic Flame Temperature	3710.5 F	Calculated Flame Temperature	2760.7 F		
Combustion Gas Mass Flow					
Carbon dioxide	19561.136 lb/h	Water	13279.915 lb/h	Oxygen	99.425 lb/
Nitrogen	80925.175 lb/h	Sulphur dioxide	0.000 lb/h	Nitrogen monoxide	9.220 lb/
Carbon monoxide	50.143 lb/h	Methane	18.973 lb/h	Argon	1330.011 lb/
Combustion Gas Mole Flow					
Carbon dioxide	0.123 lbmole/s	Water	0.205 lbmole/s	Oxygen	0.001 lbmole/
Nitrogen	0.802 lbmole/s	Sulphur dioxide	0.000 lbmole/s	Nitrogen monoxide	0.000 lbmole/
Carbon monoxide	0.000 lbmole/s	Methane	0.000 lbmole/s	Argon	0.009 lbmole/
Purge Gas Results					
Fixed velocity rate	38.471 lb/h	Fixed volume velocity	0.004 ft/s	Fixed volume rate	0.691 lb/
HUSA velocity	0.012 ft/s	Reduced HUSA velocity	0.012 ft/s	API velocity	----
HUSA rate	2.256 lb/h	Reduced HUSA rate	2.256 lb/h	API flow	----

Receptor Points

Receptor Point Summary

Name	Status	Northing ft	Easting ft	Elevation ft	Radiation btu/h/ft2	Avg. Noise dB	Noise dB	NoiseA dB
RP 1 - Walkway	Active	63.15	-82.04	4.92	448	81.3	91.3	89.9

Name	Status	Northing ft	Easting ft	Elevation ft	Radiation btu/h/ft2	Avg. Noise dB	Noise dB	NoiseA dB
RP 2 - Left Pipeline	Active	0.00	-87.01	5.00	472	81.9	91.9	90.6
RP 3 - Bottom Pipelline	Active	-90.22	0.00	5.00	482	81.8	91.8	90.4
RP 3 - Right Pipeline	Active	0.00	86.90	5.00	472	81.9	91.9	90.6

Receptor Point: RP 1 - Walkway - Active

Input Data

Location		Properties		Sizing	
Northing	63.15 ft	Emissivity	0.70	Max. Radiation	-----
Easting	-82.04 ft	Absorbtivity	0.70	Observed Radiation	-----
Elevation	4.92 ft	Mass per Area	4.9 lb/ft2	Max. Noise	-----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	-----
Angle From North	-52.41 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	-----
Angle To Horizontal	2.72 °	Initial Temp.	60.0 F	Max.Temp.	-----
Radius	103.65 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

				Noise	
Radiation	448 btu/h/ft2	Temperature	107.3 F	Average Noise	81.3 dl
Concentration	-----	Local Wind Speed	58.661 ft/s	Noise	91.3 dl
Time to TDU Limit	182.9 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	89.9 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.0 dB	21.6 dB
62.5 Hz	64.4 dB	38.2 dB
125.0 Hz	71.3 dB	55.2 dB
250.0 Hz	79.5 dB	70.9 dB
500.0 Hz	88.0 dB	84.8 dB
1000.0 Hz	85.2 dB	85.2 dB
2000.0 Hz	82.3 dB	83.5 dB
4000.0 Hz	78.8 dB	79.8 dB
8000.0 Hz	74.4 dB	73.3 dB
16000.0 Hz	66.5 dB	59.9 dB

(*) Noise A = Noise weighted for human ear sensitivity

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Noise Chart

Time Dependent Curves

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	65.9	15.9
20.0	71.1	31.7	30.0	75.6	47.6
40.0	79.6	63.4	50.0	83.0	79.3
60.0	86.1	95.1	70.0	88.7	111.0
80.0	91.1	126.8	90.0	93.1	142.7
100.0	94.9	158.5	110.0	96.4	174.4
120.0	97.8	190.2	130.0	99.0	206.1
140.0	100.1	221.9	150.0	101.0	237.8
160.0	101.8	253.6	170.0	102.5	269.5
180.0	103.1	285.4	190.0	103.6	301.2
200.0	104.1	317.1	210.0	104.5	332.9
220.0	104.8	348.8	230.0	105.2	364.6
240.0	105.4	380.5	250.0	105.7	396.3
260.0	105.9	412.2	270.0	106.1	428.0
280.0	106.2	443.9	290.0	106.4	459.7
300.0	106.5	475.6	310.0	106.6	491.4
320.0	106.7	507.3	330.0	106.8	523.1
340.0	106.8	539.0	350.0	106.9	554.8
360.0	107.0	570.7	370.0	107.0	586.6
380.0	107.0	602.4	390.0	107.1	618.3
400.0	107.1	634.1	410.0	107.1	650.0
420.0	107.2	665.8	430.0	107.2	681.7
440.0	107.2	697.5	450.0	107.2	713.4
460.0	107.2	729.2	470.0	107.2	745.1
480.0	107.3	760.9	490.0	107.3	776.8
500.0	107.3	792.6	510.0	107.3	808.5
520.0	107.3	824.3	530.0	107.3	840.2
540.0	107.3	856.1	550.0	107.3	871.9
560.0	107.3	887.8	570.0	107.3	903.6
580.0	107.3	919.5	590.0	107.3	935.3
600.0	107.3	951.2	610.0	107.3	967.0
620.0	107.3	982.9	630.0	107.3	998.7

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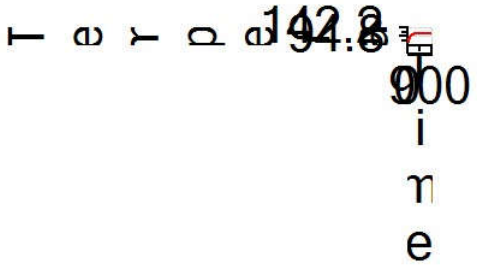
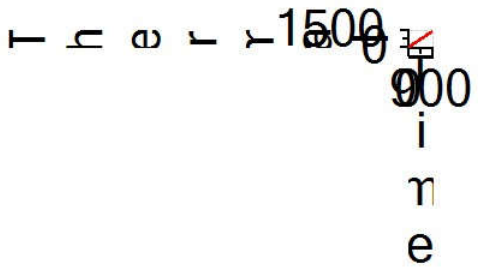
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
640.0	107.3	1014.6	650.0	107.3	1030.4
660.0	107.3	1046.3	670.0	107.3	1062.1
680.0	107.3	1078.0	690.0	107.3	1093.8
700.0	107.3	1109.7	710.0	107.3	1125.5
720.0	107.3	1141.4	730.0	107.3	1157.3
740.0	107.3	1173.1	750.0	107.3	1189.0
760.0	107.3	1204.8	770.0	107.3	1220.7
780.0	107.3	1236.5	790.0	107.3	1252.4
800.0	107.3	1268.2	810.0	107.3	1284.1
820.0	107.3	1299.9	830.0	107.3	1315.8
840.0	107.3	1331.6	850.0	107.3	1347.5
860.0	107.3	1363.3	870.0	107.3	1379.2
880.0	107.3	1395.0	890.0	107.3	1410.9
900.0	107.3	1426.8			

Temperature Chart



TDU Chart



Receptor Point: RP 2 - Left Pipeline - Active

Input Data

Location		Properties		Sizing	
Northing	0.00 ft	Emissivity	0.70	Max. Radiation	----
Easting	-87.01 ft	Absorbtivity	0.70	Observed Radiation	----
Elevation	5.00 ft	Mass per Area	4.9 lb/ft2	Max. Noise	----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	----
Angle From North	270.00 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	----
Angle To Horizontal	3.29 °	Initial Temp.	60.0 F	Max.Temp.	----
Radius	87.16 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

				Noise	
Radiation	472 btu/h/ft2	Temperature	108.0 F	Average Noise	81.9 dl
Concentration	------	Local Wind Speed	58.661 ft/s	Noise	91.9 dl
Time to TDU Limit	170.6 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	90.6 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.1 dB	21.7 dB
62.5 Hz	64.8 dB	38.6 dB
125.0 Hz	71.9 dB	55.8 dB
250.0 Hz	80.1 dB	71.5 dB
500.0 Hz	88.6 dB	85.4 dB
1000.0 Hz	85.8 dB	85.8 dB
2000.0 Hz	83.0 dB	84.2 dB
4000.0 Hz	79.5 dB	80.5 dB
8000.0 Hz	75.1 dB	74.0 dB
16000.0 Hz	67.2 dB	60.6 dB

(*) Noise A = Noise weighted for human ear sensitivity

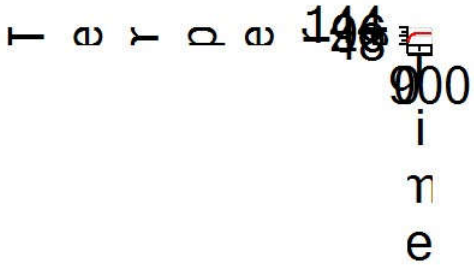
Noise Chart

Time Dependent Curves

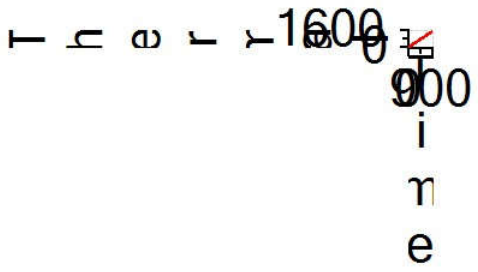
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.0	17.0
20.0	71.2	34.0	30.0	75.8	51.0
40.0	79.8	68.0	50.0	83.4	85.0
60.0	86.4	102.0	70.0	89.1	119.0
80.0	91.5	136.0	90.0	93.6	153.0
100.0	95.4	170.0	110.0	97.0	187.0
120.0	98.3	204.0	130.0	99.6	221.0
140.0	100.6	238.0	150.0	101.5	255.0
160.0	102.3	272.0	170.0	103.1	289.0
180.0	103.7	306.1	190.0	104.2	323.1
200.0	104.7	340.1	210.0	105.1	357.1
220.0	105.5	374.1	230.0	105.8	391.1
240.0	106.1	408.1	250.0	106.3	425.1
260.0	106.5	442.1	270.0	106.7	459.1
280.0	106.9	476.1	290.0	107.0	493.1
300.0	107.1	510.1	310.0	107.2	527.1

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
320.0	107.3	544.1	330.0	107.4	561.1
340.0	107.5	578.1	350.0	107.6	595.1
360.0	107.6	612.1	370.0	107.7	629.1
380.0	107.7	646.1	390.0	107.7	663.1
400.0	107.8	680.1	410.0	107.8	697.1
420.0	107.8	714.1	430.0	107.8	731.1
440.0	107.9	748.1	450.0	107.9	765.1
460.0	107.9	782.1	470.0	107.9	799.1
480.0	107.9	816.1	490.0	107.9	833.1
500.0	107.9	850.1	510.0	107.9	867.1
520.0	108.0	884.1	530.0	108.0	901.2
540.0	108.0	918.2	550.0	108.0	935.2
560.0	108.0	952.2	570.0	108.0	969.2
580.0	108.0	986.2	590.0	108.0	1003.2
600.0	108.0	1020.2	610.0	108.0	1037.2
620.0	108.0	1054.2	630.0	108.0	1071.2
640.0	108.0	1088.2	650.0	108.0	1105.2
660.0	108.0	1122.2	670.0	108.0	1139.2
680.0	108.0	1156.2	690.0	108.0	1173.2
700.0	108.0	1190.2	710.0	108.0	1207.2
720.0	108.0	1224.2	730.0	108.0	1241.2
740.0	108.0	1258.2	750.0	108.0	1275.2
760.0	108.0	1292.2	770.0	108.0	1309.2
780.0	108.0	1326.2	790.0	108.0	1343.2
800.0	108.0	1360.2	810.0	108.0	1377.2
820.0	108.0	1394.2	830.0	108.0	1411.2
840.0	108.0	1428.2	850.0	108.0	1445.2
860.0	108.0	1462.2	870.0	108.0	1479.2
880.0	108.0	1496.2	890.0	108.0	1513.3
900.0	108.0	1530.3			

Temperature Chart



TDU Chart



Receptor Point: RP 3 - Bottom Pipelline - Active

Input Data

Location		Properties		Sizing	
Northing	-90.22 ft	Emissivity	0.70	Max. Radiation	----
Easting	0.00 ft	Absorbtivity	0.70	Observed Radiation	----
Elevation	5.00 ft	Mass per Area	4.9 lb/ft2	Max. Noise	----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	----
Angle From North	180.00 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	----
Angle To Horizontal	3.17 °	Initial Temp.	60.0 F	Max.Temp.	----
Radius	90.36 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

				Noise	
Radiation	482 btu/h/ft2	Temperature	108.3 F	Average Noise	81.8 dl
Concentration	------	Local Wind Speed	58.661 ft/s	Noise	91.8 dl
Time to TDU Limit	165.7 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	90.4 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.1 dB	21.7 dB
62.5 Hz	64.7 dB	38.5 dB
125.0 Hz	71.8 dB	55.7 dB
250.0 Hz	80.0 dB	71.4 dB
500.0 Hz	88.5 dB	85.3 dB
1000.0 Hz	85.7 dB	85.7 dB
2000.0 Hz	82.9 dB	84.1 dB
4000.0 Hz	79.4 dB	80.4 dB
8000.0 Hz	75.0 dB	73.9 dB
16000.0 Hz	67.1 dB	60.5 dB

(*) Noise A = Noise weighted for human ear sensitivity

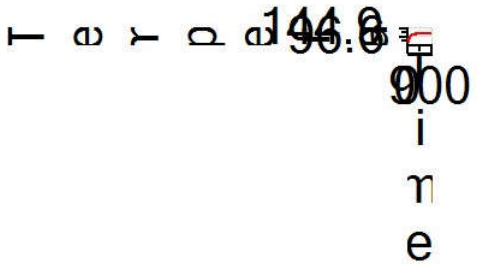
Noise Chart

Time Dependent Curves

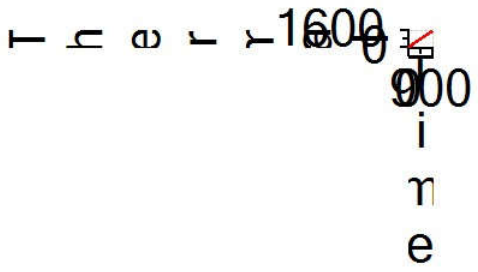
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.0	17.5
20.0	71.3	35.0	30.0	75.9	52.5
40.0	80.0	70.0	50.0	83.5	87.5
60.0	86.6	105.0	70.0	89.3	122.5
80.0	91.7	140.0	90.0	93.8	157.5
100.0	95.6	175.0	110.0	97.2	192.5
120.0	98.6	210.1	130.0	99.8	227.6
140.0	100.9	245.1	150.0	101.8	262.6
160.0	102.6	280.1	170.0	103.3	297.6
180.0	103.9	315.1	190.0	104.5	332.6
200.0	105.0	350.1	210.0	105.4	367.6
220.0	105.7	385.1	230.0	106.1	402.6
240.0	106.3	420.1	250.0	106.6	437.6
260.0	106.8	455.1	270.0	107.0	472.6
280.0	107.1	490.1	290.0	107.3	507.6
300.0	107.4	525.1	310.0	107.5	542.6

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
320.0	107.6	560.1	330.0	107.7	577.6
340.0	107.8	595.2	350.0	107.8	612.7
360.0	107.9	630.2	370.0	107.9	647.7
380.0	108.0	665.2	390.0	108.0	682.7
400.0	108.1	700.2	410.0	108.1	717.7
420.0	108.1	735.2	430.0	108.1	752.7
440.0	108.1	770.2	450.0	108.2	787.7
460.0	108.2	805.2	470.0	108.2	822.7
480.0	108.2	840.2	490.0	108.2	857.7
500.0	108.2	875.2	510.0	108.2	892.7
520.0	108.2	910.2	530.0	108.2	927.7
540.0	108.2	945.2	550.0	108.3	962.7
560.0	108.3	980.3	570.0	108.3	997.8
580.0	108.3	1015.3	590.0	108.3	1032.8
600.0	108.3	1050.3	610.0	108.3	1067.8
620.0	108.3	1085.3	630.0	108.3	1102.8
640.0	108.3	1120.3	650.0	108.3	1137.8
660.0	108.3	1155.3	670.0	108.3	1172.8
680.0	108.3	1190.3	690.0	108.3	1207.8
700.0	108.3	1225.3	710.0	108.3	1242.8
720.0	108.3	1260.3	730.0	108.3	1277.8
740.0	108.3	1295.3	750.0	108.3	1312.8
760.0	108.3	1330.3	770.0	108.3	1347.8
780.0	108.3	1365.4	790.0	108.3	1382.9
800.0	108.3	1400.4	810.0	108.3	1417.9
820.0	108.3	1435.4	830.0	108.3	1452.9
840.0	108.3	1470.4	850.0	108.3	1487.9
860.0	108.3	1505.4	870.0	108.3	1522.9
880.0	108.3	1540.4	890.0	108.3	1557.9
900.0	108.3	1575.4			

Temperature Chart



TDU Chart



Receptor Point: RP 3 - Right Pipeline - Active

Input Data

Location		Properties		Sizing	
Northing	0.00 ft	Emissivity	0.70	Max. Radiation	----
Easting	86.90 ft	Absorbtivity	0.70	Observed Radiation	----
Elevation	5.00 ft	Mass per Area	4.9 lb/ft2	Max. Noise	----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	----
Angle From North	90.00 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	----
Angle To Horizontal	3.29 °	Initial Temp.	60.0 F	Max.Temp.	----
Radius	87.04 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

				Noise	
Radiation	472 btu/h/ft2	Temperature	108.0 F	Average Noise	81.9 dl
Concentration	------	Local Wind Speed	58.661 ft/s	Noise	91.9 dl
Time to TDU Limit	170.5 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	90.6 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.1 dB	21.7 dB
62.5 Hz	64.8 dB	38.6 dB
125.0 Hz	71.9 dB	55.8 dB
250.0 Hz	80.1 dB	71.5 dB
500.0 Hz	88.6 dB	85.4 dB
1000.0 Hz	85.8 dB	85.8 dB
2000.0 Hz	83.0 dB	84.2 dB
4000.0 Hz	79.5 dB	80.5 dB
8000.0 Hz	75.1 dB	74.0 dB
16000.0 Hz	67.2 dB	60.6 dB

(*) Noise A = Noise weighted for human ear sensitivity

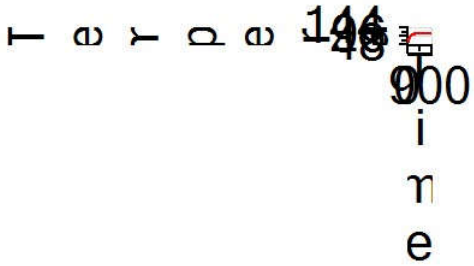
Noise Chart

Time Dependent Curves

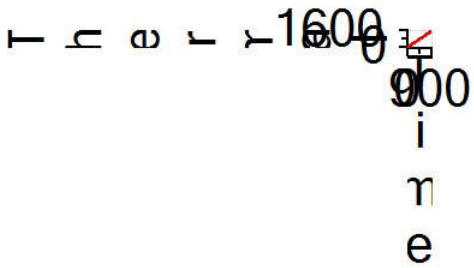
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.0	17.0
20.0	71.2	34.0	30.0	75.8	51.0
40.0	79.8	68.0	50.0	83.4	85.0
60.0	86.4	102.1	70.0	89.1	119.1
80.0	91.5	136.1	90.0	93.6	153.1
100.0	95.4	170.1	110.0	97.0	187.1
120.0	98.3	204.1	130.0	99.6	221.1
140.0	100.6	238.1	150.0	101.5	255.1
160.0	102.4	272.1	170.0	103.1	289.2
180.0	103.7	306.2	190.0	104.2	323.2
200.0	104.7	340.2	210.0	105.1	357.2
220.0	105.5	374.2	230.0	105.8	391.2
240.0	106.1	408.2	250.0	106.3	425.2
260.0	106.5	442.2	270.0	106.7	459.2
280.0	106.9	476.3	290.0	107.0	493.3
300.0	107.1	510.3	310.0	107.2	527.3

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
320.0	107.3	544.3	330.0	107.4	561.3
340.0	107.5	578.3	350.0	107.6	595.3
360.0	107.6	612.3	370.0	107.7	629.3
380.0	107.7	646.3	390.0	107.7	663.4
400.0	107.8	680.4	410.0	107.8	697.4
420.0	107.8	714.4	430.0	107.8	731.4
440.0	107.9	748.4	450.0	107.9	765.4
460.0	107.9	782.4	470.0	107.9	799.4
480.0	107.9	816.4	490.0	107.9	833.4
500.0	107.9	850.5	510.0	107.9	867.5
520.0	108.0	884.5	530.0	108.0	901.5
540.0	108.0	918.5	550.0	108.0	935.5
560.0	108.0	952.5	570.0	108.0	969.5
580.0	108.0	986.5	590.0	108.0	1003.5
600.0	108.0	1020.5	610.0	108.0	1037.5
620.0	108.0	1054.6	630.0	108.0	1071.6
640.0	108.0	1088.6	650.0	108.0	1105.6
660.0	108.0	1122.6	670.0	108.0	1139.6
680.0	108.0	1156.6	690.0	108.0	1173.6
700.0	108.0	1190.6	710.0	108.0	1207.6
720.0	108.0	1224.6	730.0	108.0	1241.7
740.0	108.0	1258.7	750.0	108.0	1275.7
760.0	108.0	1292.7	770.0	108.0	1309.7
780.0	108.0	1326.7	790.0	108.0	1343.7
800.0	108.0	1360.7	810.0	108.0	1377.7
820.0	108.0	1394.7	830.0	108.0	1411.7
840.0	108.0	1428.8	850.0	108.0	1445.8
860.0	108.0	1462.8	870.0	108.0	1479.8
880.0	108.0	1496.8	890.0	108.0	1513.8
900.0	108.0	1530.8			

Temperature Chart



TDU Chart



Receptor Grids

Receptor Grid: Grid 1 - Active

Input Data

Location		**Properties**		**Noise**	
Orientation	Northing-Easting	Emissivity	0.70	Noise Basis	Nois
Offset	0.00 ft	Point Absorbtivity	0.70		
Axis 1	Northing	Area Ratio	2.000		
Axis 2	Easting	Point Orientation	None		

Axis information

Northing	Minimum	-328.08 ft	Maximum	328.08 ft	Nbr. of Points	4
Easting	Minimum	-328.08 ft	Maximum	328.08 ft	Nbr. of Points	4

Results

Maximum Radiation

Sizing Limit	-----	Find Max. Rad.	True	Initial Grid Points	1
Radiation	528 btu/h/ft2	Northing	-11.74 ft	Easting	0.01

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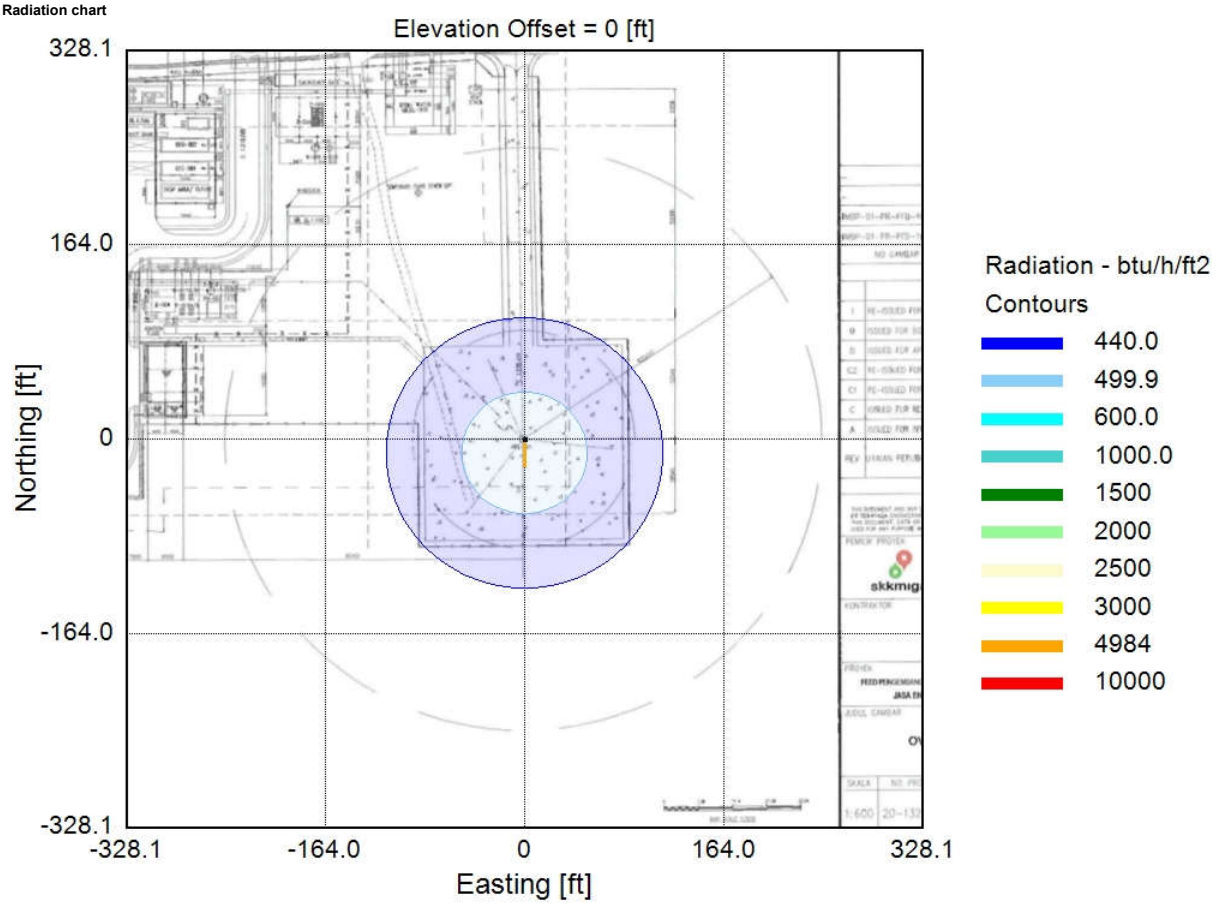
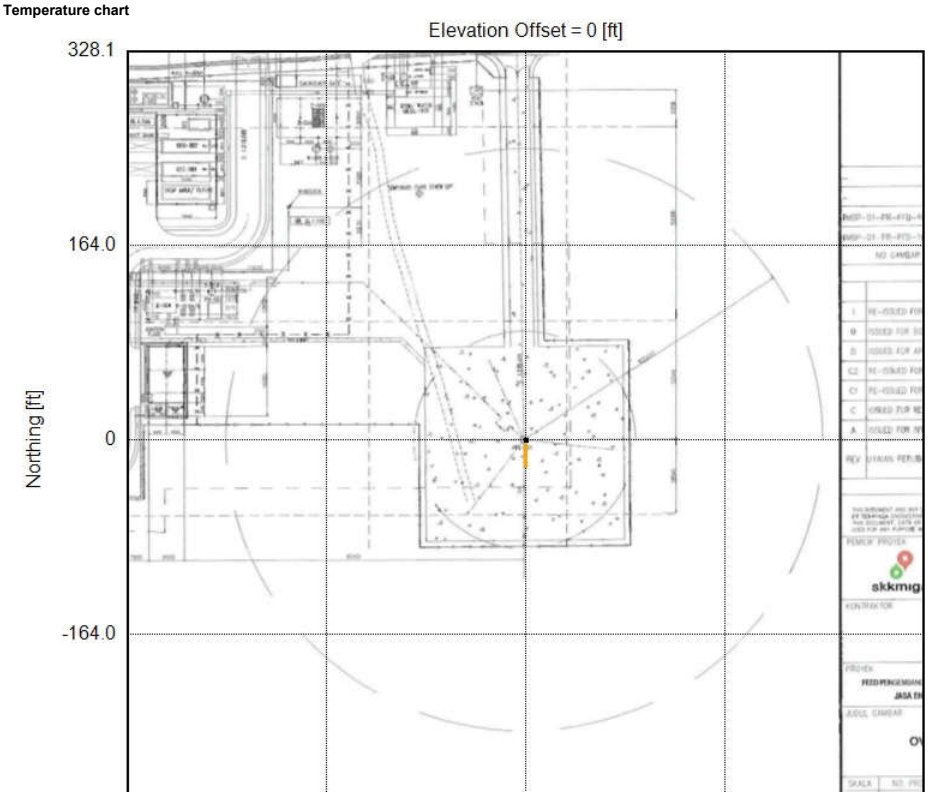
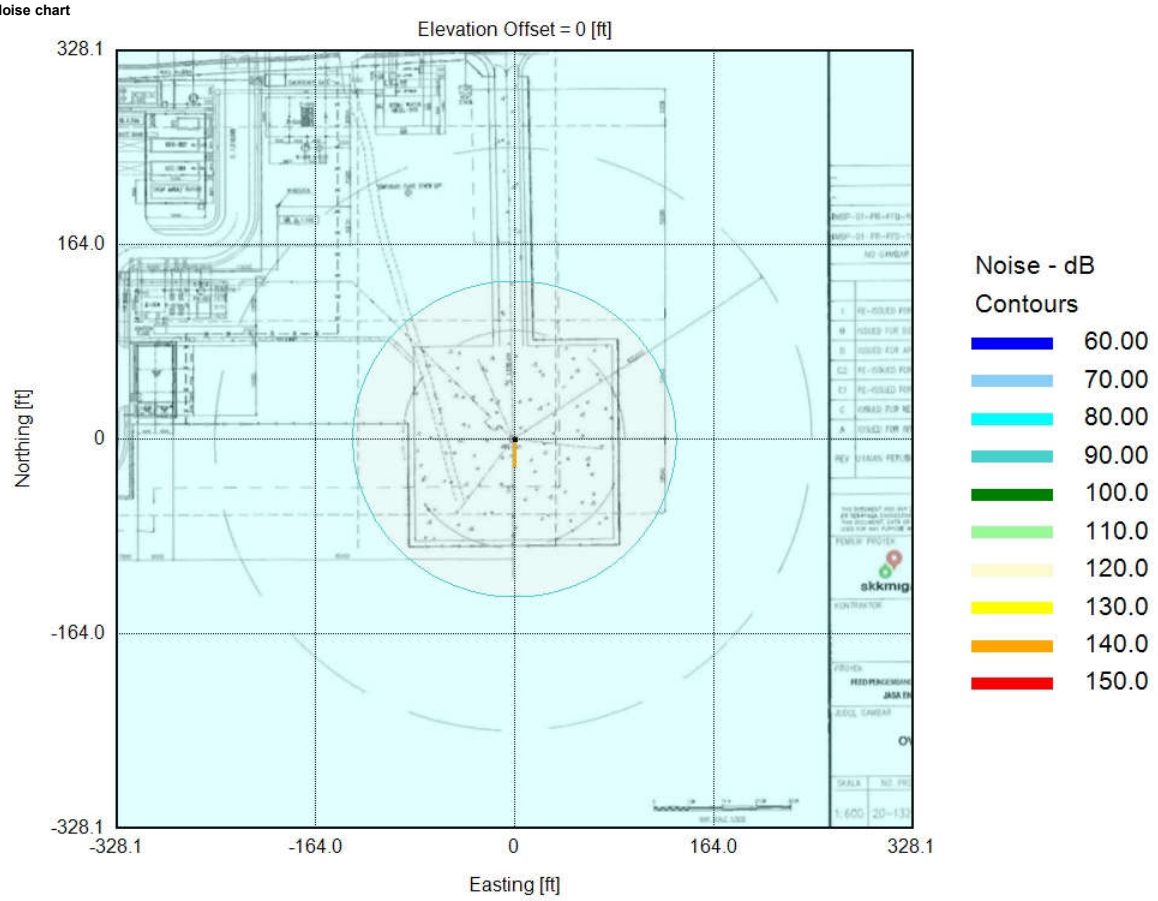
Radiation chart



Temperature chart



Noise chart



Messages

Level	Message
Info:	Calculations started at 12/24/2025 3:53:18 PM
Info:	Calculating base case
Info:	Point RP 1 - Walkway : Calculated
Info:	Point RP 2 - Left Pipeline : Calculated
Info:	Point RP 3 - Bottom Pipelline : Calculated
Info:	Point RP 3 - Right Pipeline : Calculated
Info:	Grid Grid 1 : Calculated
Info:	Calculations completed at 3:53:18 PM

Environmental data

Environment: Environment 1 - Active

Atmosphere					
Pressure	14.696 psi	Temperature	95.0 F	Humidity	87.00 %
Wind Speed	58.661 ft/s	Wind Direction	0.00 °		
Background					
BackGround Noise	60.0 dB	Solar Radiation	330 btu/h/ft2		
Inc. Background Noise	Active	Inc. Solar Radiation	Active		
Transmissivity					
Method	Calculated	Min Value	1.00	Max Value	1.0
Wind Rose Calculations					
Wind Speed Array	Ignored	Wind Speed vs Direction	Ignored		
Dispersion Data					
Stability Class	PasquillD	Terrain Class	Urban	Surface Roughness	0.82
Wind Reference Height	32.81 ft	Correct Wind Speed	False		

Stack Configuration Data

Stack: FL-001 - Active

Input Data

Base point name	FL-001 (Base)	Base refers to	Origin		
Base - Northing	0.00 ft	Base - Easting	0.00 ft	Base - Elevation	0.00
Base - Angle from North	0.00 °	Base - Angle to Horizontal	0.00 °	Base - Radius	0.00
Stack					
Rotation from North	0.00 °	Angle to Horizontal	90.00 °	Stack Length	82.00
Sterile Area					
Elevation	0.00 ft	Noise Basis	Noise	Run SA Calcs	Fals

Fluid Data

Fluid: Fluid 1

Input Data

Thermo Method	Flaresim				
Use Temp Corr.	False	Use RK ZFactor	True		
LHV	16621.01 btu/lb	Tc	151.8 F	LEL	4.05 %
Fluid Temperature	187.0 F	Pc	1174.387 psi	Saturation	100.00 %
Molecular Weight	26.254	Cp/Cv	1.266	Ref. Pressure	18.286 ps

Composition in mole fraction

Methane	0.470000	Ethane	0.080000	Propane	0.14000
i-Butane	0.020000	n-Butane	0.030000	i-Pentane	0.01000
n-Pentane	0.010000	n-Hexane	0.010000	Water	0.20000
Carbon dioxide	0.030000	Nitrogen	0.000000	Oxygen	0.00000

Results

Flue to Feed Volume Ratio	3.820	Stoichiometric O2	2.500
Enthalpy @ 25°C	192.96 btu/lb	Enthalpy @ Fluid Temp.	243.66 btu/lb

Stoichiometric Flue Gas Composition

Carbon Dioxide	0.3769634 mole/mole	Water	0.6230366 mole/mole	Oxygen	0.0000000 mole/mol
Nitrogen	0.0000000 mole/mole	Sulphur Dioxide	0.0000000 mole/mole		

Tip Data

Tip: Tip 1 - Active

Input Data

General

Tip Type	Pipe	Located on stack	FL-001	Nbr of Burners	
Angle to Horizontal	90.00 °	Angle to North	0.00 °		
Tip Length	40.00 ft	Tip Diameter	6.00 in	Tip Opening	100.00 ?
Seal Type	None	Diameter Basis	Tip		

Fluids

Fluid	Fluid 1	LP Fluid		Assist Fluid	
Fluid Mass Flow	10146.601 lb/h	LP Mass Flow	0.000 lb/h	Assist Mass Flow	0.000 lb/
Fluid Mole Flow	0.107 lbmole/s	LP Mole Flow	0.000 lbmole/s		

Pressure Drop

Riser Diameter	12.00 in	Tip Outlet Pressure	-----	Contraction Coeff.	1.00
Roughness	0.00098 in	Exit Loss	1.000		

Flame and Radiation

F Factor Method	User Specified	User F Factor	0.30	Correct User F Factor	Fals
Combustion Air Ratio	1.000	Flame Temperature	3631.7 F		
Flame Length Method	Brzustowski				
Radiation Method	Global	Flame Element Count	1	Element Position	50.00 ?

Noise

Noise Method	Acoustic Eff.	Comb. Efficiency	0.00010 %	Peak Frequency	500 H
Jet Noise Method	Flaresim				

Purge Gas

Purge Gas Fluid	Nitrogen	Purge Area Basis	Stack	Purge Fixed Velocity	0.197 ft/s
Purge Fixed Volume Flow	0.003 ft3/s	Purge HUSA O2	6.00 %	Purge HUSA Height	25.00 ft

Emission Data

NOX Basis	Mass/Heat Release	CO Basis	Mass/Heat Release	HC Basis	Mass/Heat Releas
NOX Rate	0.0680320 lb/MMBtu	CO Rate	0.3700000 lb/MMBtu	HC Rate	0.1400000 lb/MMBt

Tip: Tip 1 - Results

General

Flame Length	25.87 ft	Heat Release	168.65 MMbtu/h	Tip End - Northing	0.00
API Flame Length	51.34 ft	Frac.Heat Radiated	0.30	Tip End - Easting	0.00
Exit Temperature	187.0 F	Actual Volume Flow	50.512 ft3/s	Tip End - Elevation	122.00
Assist Fluid Mass Flow	-----	Assist Fluid Flow Ratio	-----	Wind Speed At Tip	58.661 ft/

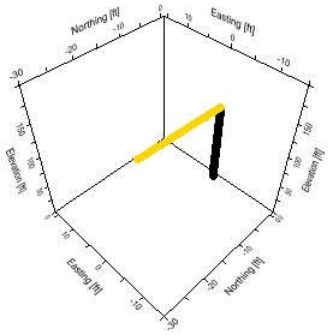
Velocities and Pressure Drop

Exit Velocity	257.256 ft/s	Tip Outlet pres.	14.696 psi	Total Outlet pres.	15.094 ps
Sonic Velocity	1242.661 ft/s	Tip Backpres.	15.157 psi	Total Backpres.	15.543 ps
Exit Mach Number	0.207	Seal Backpres.	15.306 psi	Total Backpres.	15.688 ps
Contraction Coeff.	1.000	Stack Backpres.	15.692 psi	Total Backpres.	15.715 ps
		Tip Pressure Drop	0.461 psi	Tip Total Pressure Drop	0.449 ps
		Seal Pressure Drop	0.149 psi	Seal Total Pressure Drop	0.145 ps
		Stack Pressure Drop	0.386 psi	Stack Total Pressure Drop	0.027 ps

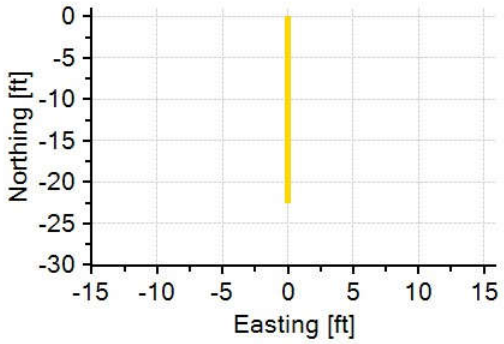
Flame Shape Points

	X	Y	Z
1	0.00 ft	0.00 ft	122.00 ft
2	-22.68 ft	0.00 ft	129.89 ft

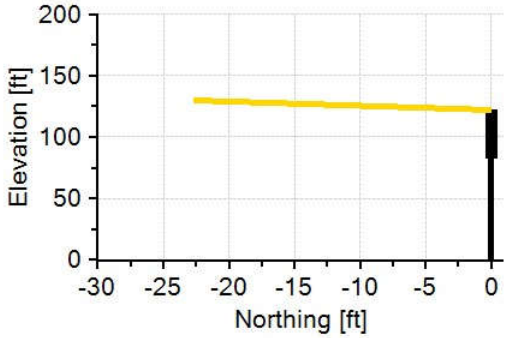
3D Flame Shape Chart



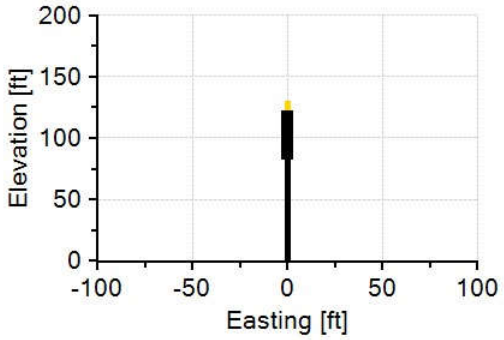
2D Flame Shape Chart, Northing-Easting



2D Flame Shape Chart, Northing-Elevation



2D Flame Shape Chart, Easting-Elevation



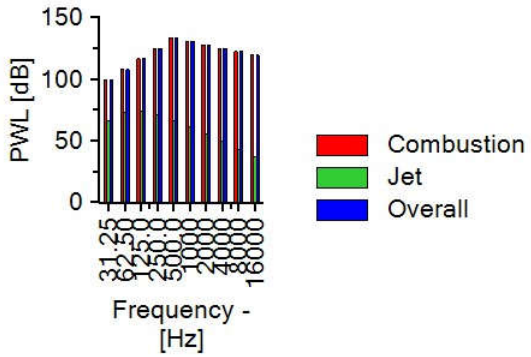
Noise Results
Peak sound pressure

Distance	98.43 ft	Sound pressure	96.4 dB

Noise Spectrum Results

Frequency	Total PWL	Combustion PWL	Jet PWL
Hz	dB	dB	dB
31.3	99.5	99.5	65.7
62.5	108.0	108.0	73.0
125.0	116.5	116.5	73.6
250.0	125.0	125.0	71.1
500.0	133.5	133.5	66.7
1000.0	130.7	130.7	61.2
2000.0	127.9	127.9	55.3
4000.0	125.1	125.1	49.3
8000.0	122.3	122.3	43.3
16000.0	119.5	119.5	37.2

Noise Spectrum



Combustion Results General

Adiabatic Flame Temperature	3710.5 F	Calculated Flame Temperature	2760.7 F

Combustion Gas Mass Flow

Carbon dioxide	24330.459 lb/h	Water	16517.774 lb/h	Oxygen	123.667 lb/
Nitrogen	100656.048 lb/h	Sulphur dioxide	0.000 lb/h	Nitrogen monoxide	11.468 lb/
Carbon monoxide	62.369 lb/h	Methane	23.599 lb/h	Argon	1654.290 lb/

Combustion Gas Mole Flow

Carbon dioxide	0.154 lbmole/s	Water	0.255 lbmole/s	Oxygen	0.001 lbmole/
Nitrogen	0.998 lbmole/s	Sulphur dioxide	0.000 lbmole/s	Nitrogen monoxide	0.000 lbmole/
Carbon monoxide	0.001 lbmole/s	Methane	0.000 lbmole/s	Argon	0.012 lbmole/

Purge Gas Results

Fixed velocity rate	38.471 lb/h	Fixed volume velocity	0.004 ft/s	Fixed volume rate	0.691 lb/
HUSA velocity	0.012 ft/s	Reduced HUSA velocity	0.012 ft/s	API velocity	----
HUSA rate	2.256 lb/h	Reduced HUSA rate	2.256 lb/h	API flow	----

Receptor Points

Receptor Point Summary

Name	Status	Northing ft	Easting ft	Elevation ft	Radiation btu/h/ft2	Avg. Noise dB	Noise dB	NoiseA dB
RP 1 - Walkway	Active	63.15	-82.04	4.92	476	82.2	92.2	90.9

Name	Status	Northing ft	Easting ft	Elevation ft	Radiation btu/h/ft2	Avg. Noise dB	Noise dB	NoiseA dB
RP 2 - Left Pipeline	Active	0.00	-87.01	5.00	505	82.8	92.8	91.5
RP 3 - Bottom Pipelline	Active	-90.22	0.00	5.00	517	82.7	92.7	91.4
RP 3 - Right Pipeline	Active	0.00	86.90	5.00	505	82.8	92.8	91.5

Receptor Point: RP 1 - Walkway - Active

Input Data

Location		Properties		Sizing	
Northing	63.15 ft	Emissivity	0.70	Max. Radiation	-----
Easting	-82.04 ft	Absorbtivity	0.70	Observed Radiation	-----
Elevation	4.92 ft	Mass per Area	4.9 lb/ft2	Max. Noise	-----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	-----
Angle From North	-52.41 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	-----
Angle To Horizontal	2.72 °	Initial Temp.	60.0 F	Max.Temp.	-----
Radius	103.65 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

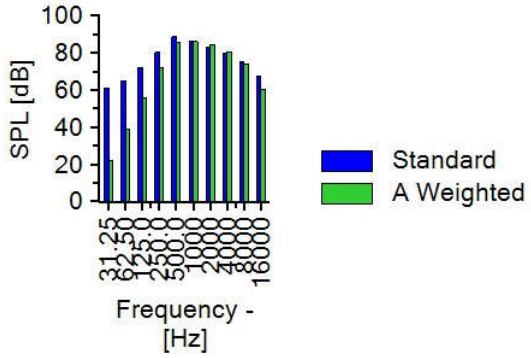
				Noise	
Radiation	476 btu/h/ft2	Temperature	108.1 F	Average Noise	82.2 dI
Concentration	-----	Local Wind Speed	58.661 ft/s	Noise	92.2 dI
Time to TDU Limit	168.7 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	90.9 dI

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.2 dB	21.8 dB
62.5 Hz	65.0 dB	38.8 dB
125.0 Hz	72.2 dB	56.1 dB
250.0 Hz	80.5 dB	71.9 dB
500.0 Hz	88.9 dB	85.7 dB
1000.0 Hz	86.1 dB	86.1 dB
2000.0 Hz	83.3 dB	84.5 dB
4000.0 Hz	79.8 dB	80.8 dB
8000.0 Hz	75.3 dB	74.2 dB
16000.0 Hz	67.3 dB	60.7 dB

(*) Noise A = Noise weighted for human ear sensitivity

Noise Chart



Time Dependent Curves

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.0	17.2
20.0	71.3	34.4	30.0	75.9	51.6
40.0	79.9	68.7	50.0	83.4	85.9
60.0	86.5	103.1	70.0	89.2	120.3
80.0	91.6	137.5	90.0	93.6	154.7
100.0	95.5	171.9	110.0	97.0	189.0
120.0	98.4	206.2	130.0	99.6	223.4
140.0	100.7	240.6	150.0	101.6	257.8
160.0	102.4	275.0	170.0	103.2	292.2
180.0	103.8	309.3	190.0	104.3	326.5
200.0	104.8	343.7	210.0	105.2	360.9
220.0	105.6	378.1	230.0	105.9	395.3
240.0	106.2	412.5	250.0	106.4	429.6
260.0	106.6	446.8	270.0	106.8	464.0
280.0	107.0	481.2	290.0	107.1	498.4
300.0	107.2	515.6	310.0	107.3	532.8
320.0	107.4	549.9	330.0	107.5	567.1
340.0	107.6	584.3	350.0	107.7	601.5
360.0	107.7	618.7	370.0	107.8	635.9
380.0	107.8	653.1	390.0	107.8	670.2
400.0	107.9	687.4	410.0	107.9	704.6
420.0	107.9	721.8	430.0	107.9	739.0
440.0	108.0	756.2	450.0	108.0	773.4
460.0	108.0	790.6	470.0	108.0	807.7
480.0	108.0	824.9	490.0	108.0	842.1
500.0	108.0	859.3	510.0	108.1	876.5
520.0	108.1	893.7	530.0	108.1	910.9
540.0	108.1	928.0	550.0	108.1	945.2
560.0	108.1	962.4	570.0	108.1	979.6
580.0	108.1	996.8	590.0	108.1	1014.0
600.0	108.1	1031.2	610.0	108.1	1048.3
620.0	108.1	1065.5	630.0	108.1	1082.7

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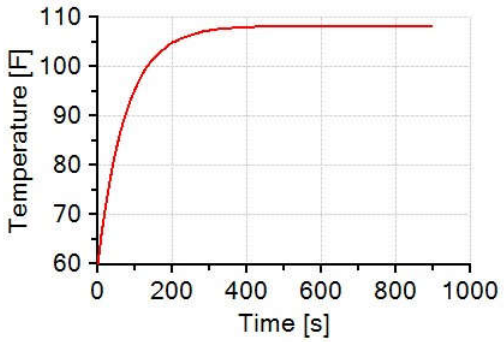
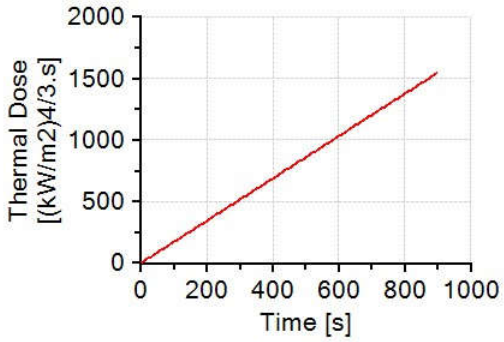
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
640.0	108.1	1099.9	650.0	108.1	1117.1
660.0	108.1	1134.3	670.0	108.1	1151.5
680.0	108.1	1168.6	690.0	108.1	1185.8
700.0	108.1	1203.0	710.0	108.1	1220.2
720.0	108.1	1237.4	730.0	108.1	1254.6
740.0	108.1	1271.8	750.0	108.1	1288.9
760.0	108.1	1306.1	770.0	108.1	1323.3
780.0	108.1	1340.5	790.0	108.1	1357.7
800.0	108.1	1374.9	810.0	108.1	1392.1
820.0	108.1	1409.2	830.0	108.1	1426.4
840.0	108.1	1443.6	850.0	108.1	1460.8
860.0	108.1	1478.0	870.0	108.1	1495.2
880.0	108.1	1512.4	890.0	108.1	1529.5
900.0	108.1	1546.7			

Temperature Chart



TDU Chart



Receptor Point: RP 2 - Left Pipeline - Active

Input Data

Location		Properties		Sizing	
Northing	0.00 ft	Emissivity	0.70	Max. Radiation	----
Easting	-87.01 ft	Absorbtivity	0.70	Observed Radiation	----
Elevation	5.00 ft	Mass per Area	4.9 lb/ft2	Max. Noise	----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	----
Angle From North	270.00 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	----
Angle To Horizontal	3.29 °	Initial Temp.	60.0 F	Max.Temp.	----
Radius	87.16 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

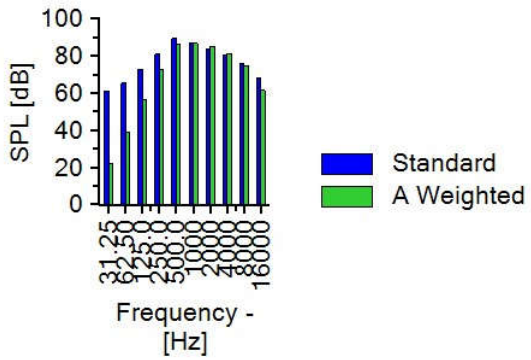
				Noise	
Radiation	505 btu/h/ft2	Temperature	108.9 F	Average Noise	82.8 dl
Concentration	-----	Local Wind Speed	58.661 ft/s	Noise	92.8 dl
Time to TDU Limit	155.8 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	91.5 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.3 dB	21.9 dB
62.5 Hz	65.5 dB	39.3 dB
125.0 Hz	72.8 dB	56.7 dB
250.0 Hz	81.1 dB	72.5 dB
500.0 Hz	89.5 dB	86.3 dB
1000.0 Hz	86.7 dB	86.7 dB
2000.0 Hz	83.9 dB	85.1 dB
4000.0 Hz	80.4 dB	81.4 dB
8000.0 Hz	76.0 dB	74.9 dB
16000.0 Hz	68.0 dB	61.4 dB

(*) Noise A = Noise weighted for human ear sensitivity

Noise Chart

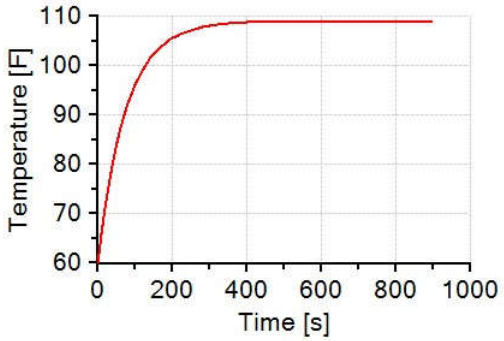


Time Dependent Curves

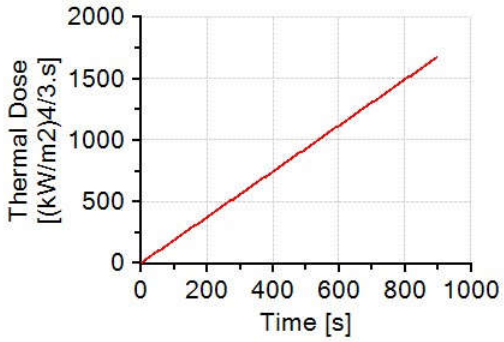
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.1	18.6
20.0	71.4	37.2	30.0	76.1	55.8
40.0	80.2	74.4	50.0	83.8	93.0
60.0	87.0	111.7	70.0	89.7	130.3
80.0	92.1	148.9	90.0	94.2	167.5
100.0	96.1	186.1	110.0	97.7	204.7
120.0	99.1	223.3	130.0	100.3	241.9
140.0	101.4	260.5	150.0	102.3	279.1
160.0	103.2	297.8	170.0	103.9	316.4
180.0	104.5	335.0	190.0	105.1	353.6
200.0	105.5	372.2	210.0	106.0	390.8
220.0	106.3	409.4	230.0	106.7	428.0
240.0	106.9	446.6	250.0	107.2	465.2
260.0	107.4	483.9	270.0	107.6	502.5
280.0	107.8	521.1	290.0	107.9	539.7
300.0	108.0	558.3	310.0	108.1	576.9

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
320.0	108.2	595.5	330.0	108.3	614.1
340.0	108.4	632.7	350.0	108.5	651.3
360.0	108.5	670.0	370.0	108.6	688.6
380.0	108.6	707.2	390.0	108.6	725.8
400.0	108.7	744.4	410.0	108.7	763.0
420.0	108.7	781.6	430.0	108.7	800.2
440.0	108.8	818.8	450.0	108.8	837.4
460.0	108.8	856.1	470.0	108.8	874.7
480.0	108.8	893.3	490.0	108.8	911.9
500.0	108.8	930.5	510.0	108.9	949.1
520.0	108.9	967.7	530.0	108.9	986.3
540.0	108.9	1004.9	550.0	108.9	1023.5
560.0	108.9	1042.2	570.0	108.9	1060.8
580.0	108.9	1079.4	590.0	108.9	1098.0
600.0	108.9	1116.6	610.0	108.9	1135.2
620.0	108.9	1153.8	630.0	108.9	1172.4
640.0	108.9	1191.0	650.0	108.9	1209.6
660.0	108.9	1228.3	670.0	108.9	1246.9
680.0	108.9	1265.5	690.0	108.9	1284.1
700.0	108.9	1302.7	710.0	108.9	1321.3
720.0	108.9	1339.9	730.0	108.9	1358.5
740.0	108.9	1377.1	750.0	108.9	1395.7
760.0	108.9	1414.4	770.0	108.9	1433.0
780.0	108.9	1451.6	790.0	108.9	1470.2
800.0	108.9	1488.8	810.0	108.9	1507.4
820.0	108.9	1526.0	830.0	108.9	1544.6
840.0	108.9	1563.2	850.0	108.9	1581.8
860.0	108.9	1600.5	870.0	108.9	1619.1
880.0	108.9	1637.7	890.0	108.9	1656.3
900.0	108.9	1674.9			

Temperature Chart



TDU Chart



Receptor Point: RP 3 - Bottom Pipelline - Active

Input Data

Location		Properties		Sizing	
Northing	-90.22 ft	Emissivity	0.70	Max. Radiation	----
Easting	0.00 ft	Absorbtivity	0.70	Observed Radiation	----
Elevation	5.00 ft	Mass per Area	4.9 lb/ft2	Max. Noise	----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	----
Angle From North	180.00 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	----
Angle To Horizontal	3.17 °	Initial Temp.	60.0 F	Max.Temp.	----
Radius	90.36 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

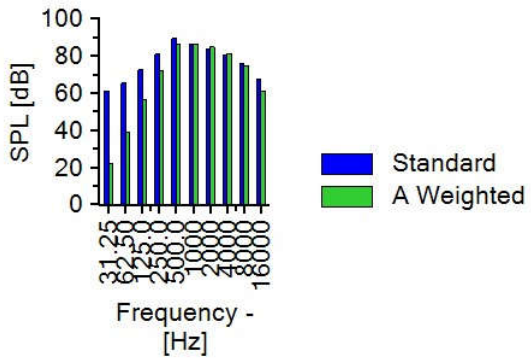
				Noise	
Radiation	517 btu/h/ft2	Temperature	109.2 F	Average Noise	82.7 dl
Concentration	-----	Local Wind Speed	58.661 ft/s	Noise	92.7 dl
Time to TDU Limit	151.0 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	91.4 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.3 dB	21.9 dB
62.5 Hz	65.4 dB	39.2 dB
125.0 Hz	72.6 dB	56.5 dB
250.0 Hz	80.9 dB	72.3 dB
500.0 Hz	89.4 dB	86.2 dB
1000.0 Hz	86.6 dB	86.6 dB
2000.0 Hz	83.8 dB	85.0 dB
4000.0 Hz	80.3 dB	81.3 dB
8000.0 Hz	75.9 dB	74.8 dB
16000.0 Hz	67.9 dB	61.3 dB

(*) Noise A = Noise weighted for human ear sensitivity

Noise Chart

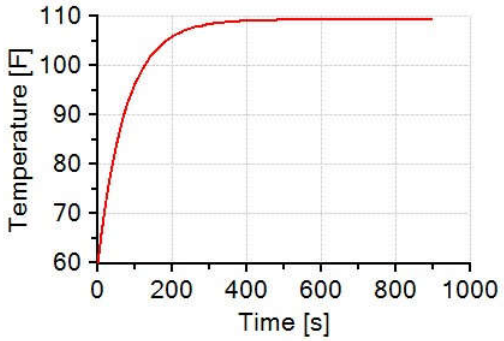


Time Dependent Curves

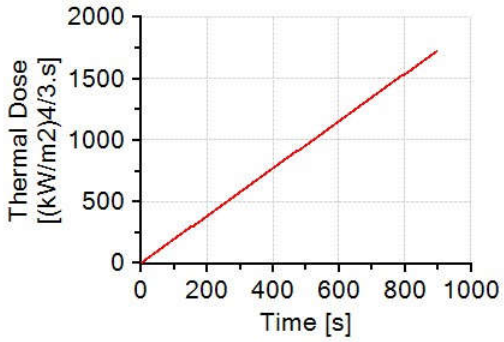
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.1	19.2
20.0	71.5	38.4	30.0	76.2	57.6
40.0	80.4	76.8	50.0	84.0	96.0
60.0	87.1	115.2	70.0	89.9	134.4
80.0	92.3	153.6	90.0	94.4	172.8
100.0	96.3	192.1	110.0	97.9	211.3
120.0	99.3	230.5	130.0	100.6	249.7
140.0	101.7	268.9	150.0	102.6	288.1
160.0	103.4	307.3	170.0	104.2	326.5
180.0	104.8	345.7	190.0	105.4	364.9
200.0	105.9	384.1	210.0	106.3	403.3
220.0	106.6	422.5	230.0	107.0	441.7
240.0	107.3	460.9	250.0	107.5	480.1
260.0	107.7	499.3	270.0	107.9	518.5
280.0	108.1	537.8	290.0	108.2	557.0
300.0	108.4	576.2	310.0	108.5	595.4

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
320.0	108.6	614.6	330.0	108.6	633.8
340.0	108.7	653.0	350.0	108.8	672.2
360.0	108.8	691.4	370.0	108.9	710.6
380.0	108.9	729.8	390.0	109.0	749.0
400.0	109.0	768.2	410.0	109.0	787.4
420.0	109.1	806.6	430.0	109.1	825.8
440.0	109.1	845.0	450.0	109.1	864.2
460.0	109.1	883.4	470.0	109.1	902.7
480.0	109.2	921.9	490.0	109.2	941.1
500.0	109.2	960.3	510.0	109.2	979.5
520.0	109.2	998.7	530.0	109.2	1017.9
540.0	109.2	1037.1	550.0	109.2	1056.3
560.0	109.2	1075.5	570.0	109.2	1094.7
580.0	109.2	1113.9	590.0	109.2	1133.1
600.0	109.2	1152.3	610.0	109.2	1171.5
620.0	109.2	1190.7	630.0	109.2	1209.9
640.0	109.2	1229.1	650.0	109.2	1248.3
660.0	109.2	1267.6	670.0	109.2	1286.8
680.0	109.2	1306.0	690.0	109.2	1325.2
700.0	109.2	1344.4	710.0	109.2	1363.6
720.0	109.2	1382.8	730.0	109.2	1402.0
740.0	109.2	1421.2	750.0	109.2	1440.4
760.0	109.2	1459.6	770.0	109.2	1478.8
780.0	109.2	1498.0	790.0	109.2	1517.2
800.0	109.2	1536.4	810.0	109.2	1555.6
820.0	109.2	1574.8	830.0	109.2	1594.0
840.0	109.2	1613.3	850.0	109.2	1632.5
860.0	109.2	1651.7	870.0	109.2	1670.9
880.0	109.2	1690.1	890.0	109.2	1709.3
900.0	109.2	1728.5			

Temperature Chart



TDU Chart



Receptor Point: RP 3 - Right Pipeline - Active

Input Data

Location		Properties		Sizing	
Northing	0.00 ft	Emissivity	0.70	Max. Radiation	----
Easting	86.90 ft	Absorbtivity	0.70	Observed Radiation	----
Elevation	5.00 ft	Mass per Area	4.9 lb/ft2	Max. Noise	----
Location rel. to	FL-001 (Base)	Area Ratio	2.000	Max. Average Noise	----
Angle From North	90.00 °	Mass Cp	0.107 btu/lb/F	Max.A Noise	----
Angle To Horizontal	3.29 °	Initial Temp.	60.0 F	Max.Temp.	----
Radius	87.04 ft	Orientation	None		
		Local Environment	Global		
		Threshold TDU	290		
		Threshold Temperature	373.15		

Results

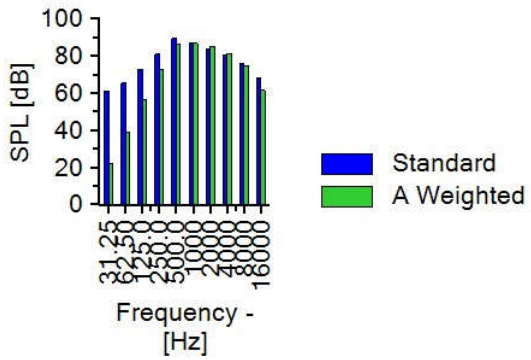
				Noise	
Radiation	505 btu/h/ft2	Temperature	108.9 F	Average Noise	82.8 dl
Concentration	-----	Local Wind Speed	58.661 ft/s	Noise	92.8 dl
Time to TDU Limit	155.8 s	Time to Temp. Limit	Limit Not Reached	Noise A (*)	91.5 dl

Noise Spectrum Results

Frequency	Noise	Noise A (*)
31.3 Hz	61.3 dB	21.9 dB
62.5 Hz	65.5 dB	39.3 dB
125.0 Hz	72.8 dB	56.7 dB
250.0 Hz	81.1 dB	72.5 dB
500.0 Hz	89.5 dB	86.3 dB
1000.0 Hz	86.7 dB	86.7 dB
2000.0 Hz	83.9 dB	85.1 dB
4000.0 Hz	80.4 dB	81.4 dB
8000.0 Hz	76.0 dB	74.9 dB
16000.0 Hz	68.0 dB	61.4 dB

(*) Noise A = Noise weighted for human ear sensitivity

Noise Chart

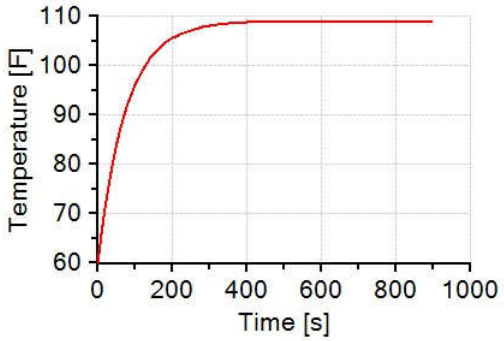


Time Dependent Curves

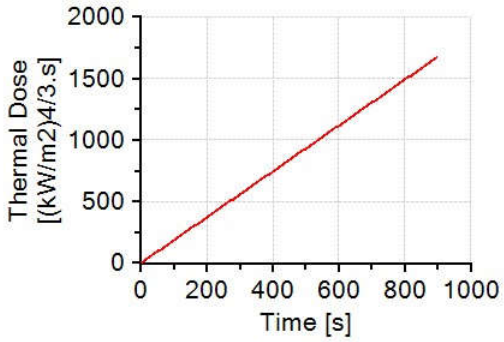
Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
0.0	60.0	0.0	10.0	66.1	18.6
20.0	71.4	37.2	30.0	76.1	55.9
40.0	80.2	74.5	50.0	83.8	93.1
60.0	87.0	111.7	70.0	89.7	130.3
80.0	92.1	148.9	90.0	94.2	167.6
100.0	96.1	186.2	110.0	97.7	204.8
120.0	99.1	223.4	130.0	100.3	242.0
140.0	101.4	260.6	150.0	102.3	279.3
160.0	103.2	297.9	170.0	103.9	316.5
180.0	104.5	335.1	190.0	105.1	353.7
200.0	105.5	372.4	210.0	106.0	391.0
220.0	106.3	409.6	230.0	106.7	428.2
240.0	106.9	446.8	250.0	107.2	465.4
260.0	107.4	484.1	270.0	107.6	502.7
280.0	107.8	521.3	290.0	107.9	539.9
300.0	108.0	558.5	310.0	108.1	577.1

Time s	Temp. F	TDU (kW/m2)4/3.s	Time s	Temp. F	TDU (kW/m2)4/3.s
320.0	108.2	595.8	330.0	108.3	614.4
340.0	108.4	633.0	350.0	108.5	651.6
360.0	108.5	670.2	370.0	108.6	688.9
380.0	108.6	707.5	390.0	108.6	726.1
400.0	108.7	744.7	410.0	108.7	763.3
420.0	108.7	781.9	430.0	108.8	800.6
440.0	108.8	819.2	450.0	108.8	837.8
460.0	108.8	856.4	470.0	108.8	875.0
480.0	108.8	893.6	490.0	108.8	912.3
500.0	108.8	930.9	510.0	108.9	949.5
520.0	108.9	968.1	530.0	108.9	986.7
540.0	108.9	1005.4	550.0	108.9	1024.0
560.0	108.9	1042.6	570.0	108.9	1061.2
580.0	108.9	1079.8	590.0	108.9	1098.4
600.0	108.9	1117.1	610.0	108.9	1135.7
620.0	108.9	1154.3	630.0	108.9	1172.9
640.0	108.9	1191.5	650.0	108.9	1210.2
660.0	108.9	1228.8	670.0	108.9	1247.4
680.0	108.9	1266.0	690.0	108.9	1284.6
700.0	108.9	1303.2	710.0	108.9	1321.9
720.0	108.9	1340.5	730.0	108.9	1359.1
740.0	108.9	1377.7	750.0	108.9	1396.3
760.0	108.9	1414.9	770.0	108.9	1433.6
780.0	108.9	1452.2	790.0	108.9	1470.8
800.0	108.9	1489.4	810.0	108.9	1508.0
820.0	108.9	1526.7	830.0	108.9	1545.3
840.0	108.9	1563.9	850.0	108.9	1582.5
860.0	108.9	1601.1	870.0	108.9	1619.7
880.0	108.9	1638.4	890.0	108.9	1657.0
900.0	108.9	1675.6			

Temperature Chart



TDU Chart



Receptor Grids

Receptor Grid: Grid 1 - Active

Input Data

Location		**Properties**		**Noise**	
Orientation	Northing-Easting	Emissivity	0.70	Noise Basis	Nois
Offset	0.00 ft	Point Absorbtivity	0.70		
Axis 1	Northing	Area Ratio	2.000		
Axis 2	Easting	Point Orientation	None		

Axis information

Northing	Minimum	-328.08 ft	Maximum	328.08 ft	Nbr. of Points	4
Easting	Minimum	-328.08 ft	Maximum	328.08 ft	Nbr. of Points	4

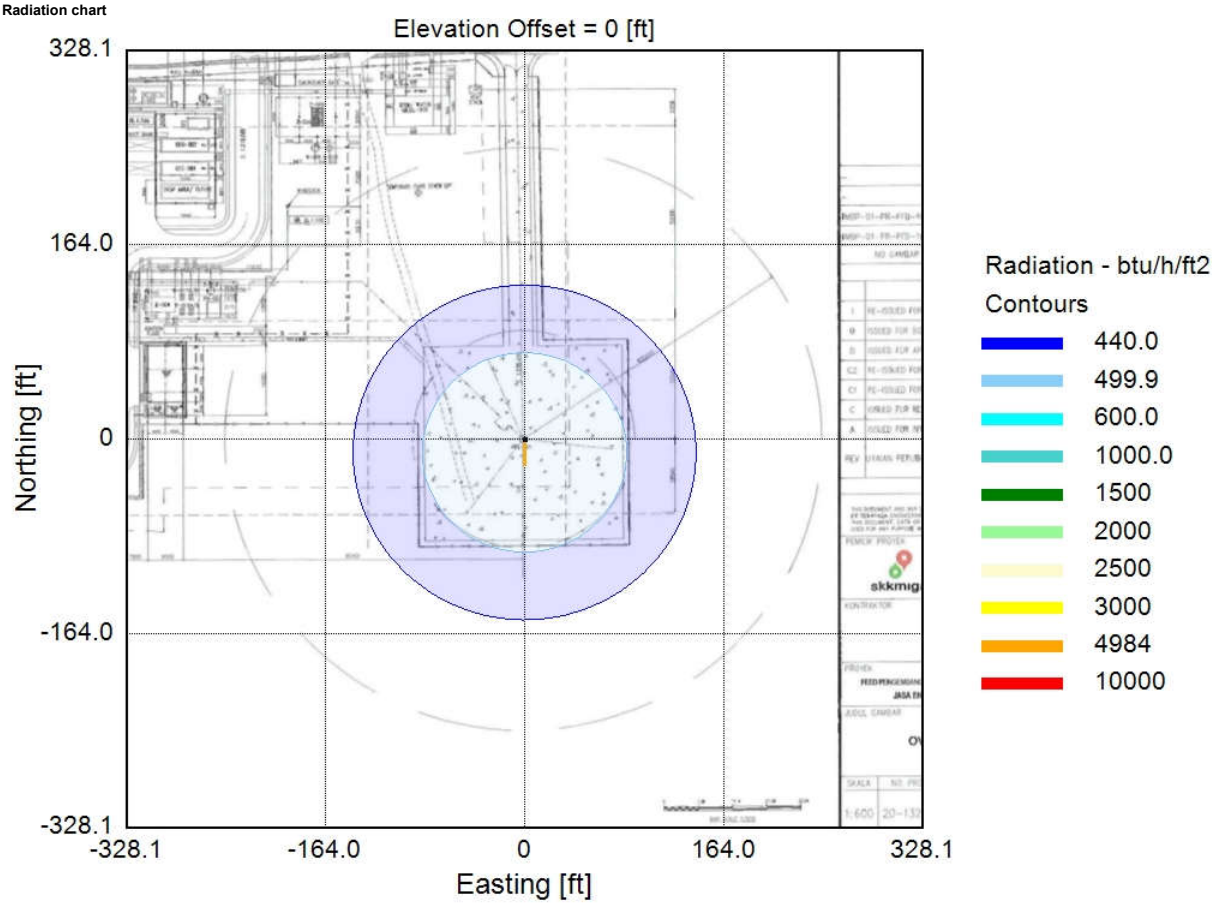
Results

Maximum Radiation

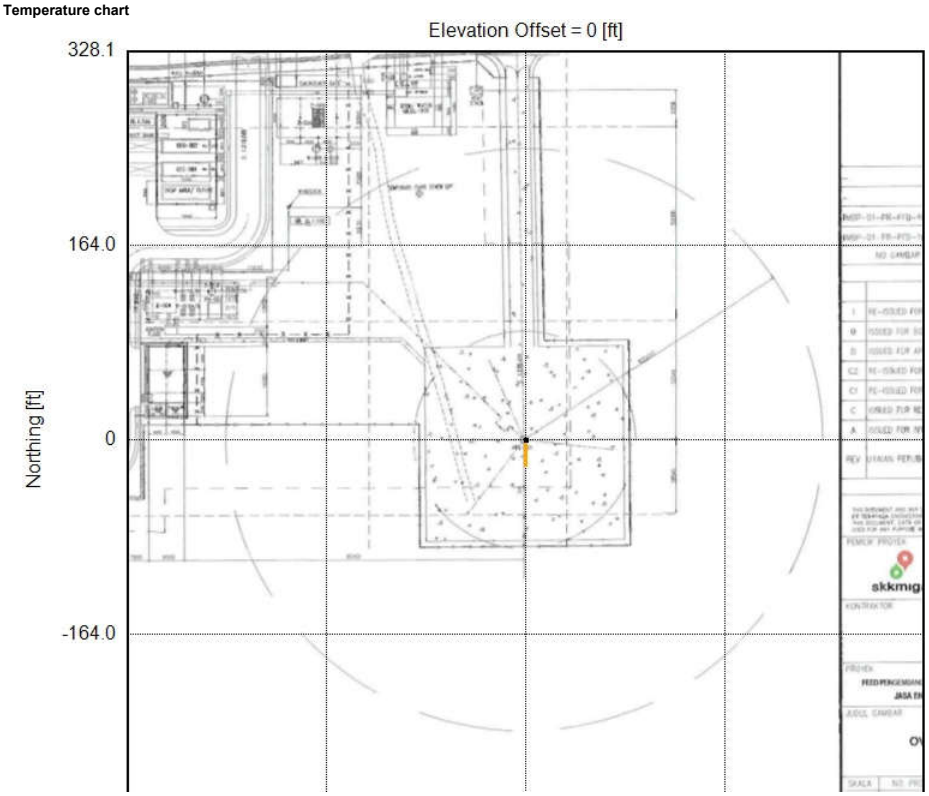
Sizing Limit	-----	Find Max. Rad.	True	Initial Grid Points	1
Radiation	574 btu/h/ft2	Northing	-11.35 ft	Easting	0.01

Page 20

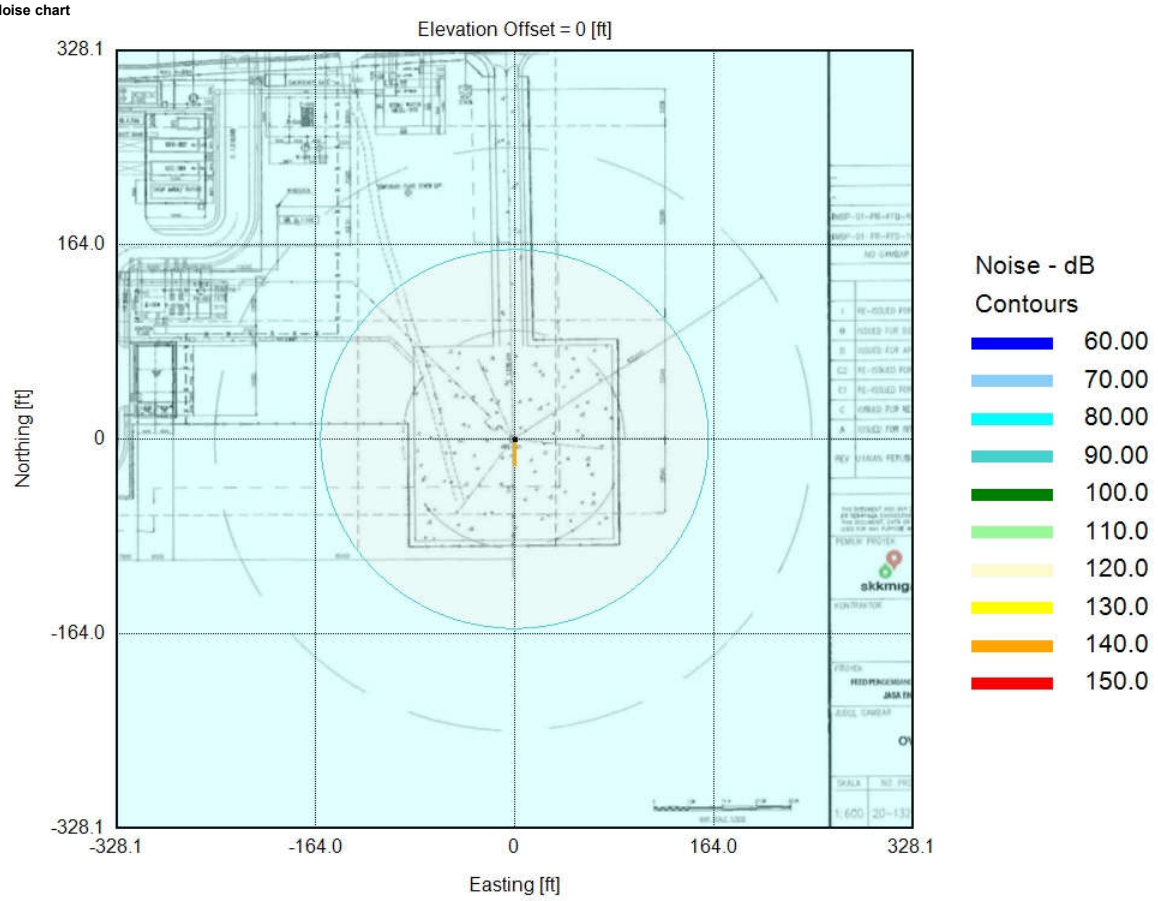
Radiation chart



Temperature chart



Noise chart



Messages

Level	Message
Info:	Calculations started at 12/24/2025 3:54:23 PM
Info:	Calculating base case
Info:	Point RP 1 - Walkway : Calculated
Info:	Point RP 2 - Left Pipeline : Calculated
Info:	Point RP 3 - Bottom Pipelline : Calculated
Info:	Point RP 3 - Right Pipeline : Calculated
Info:	Grid Grid 1 : Calculated
Info:	Calculations completed at 3:54:23 PM